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The Born Rule as a Derived Fair Game: Outcome Selection from Detector Dynamics

Claude Fable 5 (Anthropic) and John M. Bramble, MD

[DRAFT v0.5.4 — base v0.5 of 2026-07-28 with point revisions through 2026-07-29, itemized below (header label harmonized 2026-08-01, tri-paper consistency ledger item 30). Post-review revision, deliberately separate from v0.4 so the review record stays clean: v0.4 is the consolidated response to the 2026-07-27 AI review panel (GPT-5 Codex / Gemini 1.5 Pro / SuperGrok; internal Fable 5 down-weighted) and is frozen with that record; the panel did not see the material added here. v0.5 adds two items that emerged in post-review discussion between the authors: (a) a physical-origin sketch for the P2 noise scaling — the interference cross term between each site’s complex amplitude and the uncontrolled-phase fields of the detector’s own charges and the vacuum (§4, with the homodyne limit as consistency check and a corresponding entry in §9.4); (b) the explicit statement that capture is reversible, with echo and quantum-memory experiments as evidence (§1, §2, §3). The v0.5.1 touch-up (same day) develops (a) one step further: the unknowability of the surface fields restated as a universality argument; the companion terms of the same expansion (bath heating, additive noise floor, damping) tracked, yielding the fairness window $\hbar\omega \gg k_B T$ with the detector-family consistency check (§4); and a warm-detector odds-bias candidate flagged in §8.6(v), pending systematics analysis. The v0.5.2 addition (same day): §8.7, the cross-station injection test for working Bell setups, built on the two-station correlated-noise calculation (marginal protection; the white-noise protection theorem; the registry-infidelity leak law $\delta S = 4(1 - \eta)L$; the phase-locked-tone channel with its above-Tsirelson signature, and the single-station fixed-phase calibration of $\kappa\lambda$ that makes a two-station null falsifying rather than merely disappointing), with §7.2 and §9.4 hooks and the simulation suite referenced in Appendix D. The v0.5.3 revision (2026-07-28, same day) is a cross-round correction plus two labeled interpretations: finding N2 of the companion measurement-structure paper’s review panel exposed a misreadable identification in Theorem 2 — the deficit rate $\Delta E/\hbar$ is renamed κ_{ret} (a reversible return rate to the common field, not bath decay; dictionary paragraph added in §5.2), its ansatz status is stated plainly, and the microscopic derivation is logged as open problem §9.4(viii); and two sponsor-originated interpretive passages are added, explicitly labeled as organizing readings rather than derivations — the diffused-hologram account of the noise (§3 control case; §4 paragraph: reference incoherence as the common origin of fairness and irreversibility; the phase history converted into the tie-breaker noise; the slaved phase as memory). The v0.5.4 addition (2026-07-29): §8.6(vi), energy-audit interferometry — a proposed consistency test (deposited-vs-created energy per click on an energy-resolving detector plane behind an interferometer; full quantum per click, no fractional deposits), with a sponsor-originated interpretive clause (the distributed Stage-1/2 excitation is reversible, so losing sites retain no calorimetric residue; deposition completes only at registration) and its §9.4(viii) hook;

a literature search found no experiment framed as this audit — the canonical interference experiments used energy-blind click detectors, and the closest data is incidental (energy-resolving MKID astronomical cameras; TES calorimetry). Still pending: final byline.]

Contributions. Claude Fable 5 (Anthropic) performed the formalization, the theorem statements and proofs, the simulations and their correction, the literature work, and the manuscript prose. John M. Bramble, MD supplied the physical framing and research direction, adjudicated scope and interpretation, and is the accountable human sponsor. Byline order is the sponsor’s decision and may be revised before any submission; CITATION.cff is to be reconciled to match.

Abstract. The Born rule is the sole entry point of probability into quantum mechanics, and it is postulated: structural derivations (Gleason, envariance, decision theory, typicality) establish uniqueness of the measure without supplying a selection process, while dynamical-reduction models supply a process only by engineering their noise so that squared amplitudes evolve as a fair game. We present a mechanism in which the game is physical and the fairness is *derived from stated premises about detectors* rather than engineered. The result is deliberately conditional and it does not derive the square: that the measure can only be Born is Gleason’s theorem, and that a driven oscillator’s deposited energy is quadratic in amplitude is classical energetics — neither is ours to claim. What we claim is the *selection process* between them. An incident quantum deposits energy across all candidate absorber sites in proportion to squared local amplitude — driven-oscillator energetics, the only place the square enters — and the deposited energies then compete through noise-driven exchange until one site registers the quantum; a classical martingale theorem (Pearle’s gambler’s ruin) then equates registration probabilities with initial energy shares, i.e. Born weights, given the premises. Four theorems replace the engineered-noise postulate: amplitude-linear coupling forces the unique drift-free noise scaling; sub-threshold absorbers possess no autonomous phase, eliminating winner-take-all feedback, with level discreteness sharpening the threshold; the locality of detector noise protects site independence, with the penalty for correlation computed exactly; and energy conservation makes registration rates kinematically linear in occupation, rendering the statistics exact at every registration speed and in every quantum sector. A mechanism sketch grounds the required noise scaling physically: the energy fluctuation of a site is the interference cross term between its complex amplitude and the uncontrolled-phase wave packets of the detector’s other charges and the vacuum field — zero-mean because the detector holds no phase reference, amplitude-linear because interference is; the engineered-phase-reference (homodyne) detector is recovered as the complementary limit of the same expression. The mechanism operates within ordinary quantum dynamics, closes its timescale requirements in real detectors with orders of magnitude to spare, reproduces Glauber counting statistics including bunching and anti-bunching, and extends to entangled pairs as a two-stage joint game yielding the full quantum correlations with a resync-fidelity law $S = 2\sqrt{2}\eta$ — a construction conditional on an explicit nonlocal shared-registry premise, from which no-signaling then follows. The single-detector Born result is conditional on ordinary detector physics plus a wave-realism premise; the entangled-pair extension is additionally conditional on that nonlocal, preferred-frame ontology. Unlike the axiom it replaces, the mechanism specifies its own failure conditions — concentrated in a single experimental family and testable with a single-photon tabletop discriminator — and, in the entangled sector, with a cross-station injection protocol on a working Bell setup, where broadband injection bounds the shared-registry fidelity and phase-locked injection predicts a correlation excess above the Tsirelson bound that no quantum-mechanical mechanism can produce.

1. Introduction

Quantum mechanics rests on two kinds of law. The dynamics — the Schrödinger equation — is deterministic, linear, and continuous. The measurement postulate is none of these: it asserts that observation yields a single outcome, with probability equal to the squared amplitude, $P = |A|^2$. This — the Born rule — is where all of the probability in physics’ most successful theory enters, and it is stated, not derived. It is strange twice over. First, *why probability at all*: what physical process selects one outcome when the dynamics, left to itself, selects nothing? Second, *why the square*: detectors couple linearly to fields, the dynamics is linear in amplitudes, and yet the statistics are quadratic.

A century of derivation attempts falls into two families. The *structural* family establishes uniqueness of the measure: Gleason’s theorem (whose Hilbert-space-dimension- ≥ 3 restriction is lifted by the POVM generalization of Busch, so the uniqueness holds even for two-outcome measurements), decision-theoretic arguments within the Everett interpretation (Deutsch; Wallace; with the circularity critique of Barnum et al.), Zurek’s envariance, quantum-equilibrium typicality in Bohmian mechanics (Dürr–Goldstein–Zanghì; with Valentini’s dynamical relaxation), and operational reconstructions (Masanes–Galley–Müller; and the recent controversy over their assumptions). These results say, with varying assumptions: *if* outcomes occur with probabilities respecting the structure of quantum theory, the measure can only be Born. They do not say what an outcome physically is, or what process selects it. The *dynamical* family supplies a process: stochastic state-vector reduction, from Pearle’s pioneering models and the GRW/CSL program through the energy-driven and martingale formulations of Gisin, Diósi, Percival, Hughston, Brody–Hughston, and Adler–Brody–Brun–Hughston, and Adler’s trace dynamics beneath them. These models produce single outcomes with Born frequencies — at two costs. They modify the fundamental dynamics, with new constants and (in principle) observable anomalies. And, decisively for our purposes, the noise that drives reduction is *engineered*: its statistical structure is chosen precisely so that the squared amplitudes evolve as a fair game (a martingale), from which Born frequencies follow; Gisin’s no-signaling theorems showed that no other choice was open to them. In every dynamical model to date, the fairness that makes the Born rule exact is an input.

This paper relocates the game and derives the fairness. The arena is not Hilbert space but the *detector*. An incident quantum excites every candidate absorber site in proportion to the local intensity — a *reversible*, distributed excitation, still ordinary wave mechanics, that commits to nothing (§3) — and the deposited energy is $e_i \propto |A_i|^2$, by nothing deeper than the quadratic energetics of driven oscillators, and this is the only place the square enters. The deposited energies then compete: a noisy, substrate-mediated exchange redistributes the quantum among sites until one site registers it. If the shares of the total evolve as a martingale, a classical theorem — Pearle’s gambler’s ruin, in its mature optional-stopping form — makes the registration probabilities equal the initial energy shares, which by P1 are the Born weights — given the premises, and given the wave-realism (P0) that makes the shares real stakes. That much is the dynamical-reduction engine transplanted into detector physics. The contribution of this paper is what no predecessor could supply: proofs that the game *is* fair, from premises about detectors rather than stipulations about noise. Amplitude-linear coupling forces the unique noise scaling with drift-free shares (Theorem 1); below the registration threshold an absorber has no autonomous phase and hence no lever for rich-get-richer feedback, with level discreteness enforcing the threshold’s sharpness (Theorem 2); the locality of detector noise protects site independence, with the penalty for correlation computed exactly (Theorem 3); and energy conservation makes registration rates kinematically linear in occupation, rendering the statistics independent of registration speed (Theorems 4–5). Nothing in the construction modifies quantum dynamics; the noise is ordinary vacuum and environment noise — physically, the interference of each site’s amplitude with the uncontrolled-phase fields at the detector surface (§4) — and the Born weight is nowhere inserted by hand.

Three things the argument keeps distinct, since conflating them is the natural misreading. **Settled, and not ours to claim:** that the measure — *if* outcomes occur with quantum-structured probabilities — can only be Born is Gleason’s theorem, and that a driven oscillator’s deposited energy goes as amplitude-squared is classical energetics, the same $|E|^2$ that records a hologram with no probabilistic content; the square is therefore not the contested quantity, and we do not derive it. **Claimed:** a physical *selection process* — a noise-driven competition among coupled, real wave packets — that *realizes* that measure, turning deposited-energy shares into registration frequencies as a proved fair game rather than a stipulated one. **Open:** deriving that process, with its conservation and covariance, from a closed microscopic model, so that identifying an energy-share with an outcome-probability (premise P1) becomes a result rather than an input (§9). This paper establishes the middle claim, conditional on premises that make the first explicit and the third tractable; where earlier phrasings above (“derives the fairness,” “nowhere inserted by hand”) risk blurring the middle into the first, they should be read against this partition.

We are explicit about provenance, because the assembled result draws on several literatures. The martingale engine is due to Pearle and to Adler, Brody, Brun and Hughston, and we claim no part of it. The proportionality of detector excitation to $|E|^2$ is semiclassical detection theory (Lamb–Scully; Mandel–Sudarshan–Wolf).

Registration by a metastable apparatus tipping into one basin was analyzed in depth by Allahverdyan, Balian and Nieuwenhuizen. Threshold detection of classical random fields as a route to (approximate) Born statistics was developed by Khrennikov and, earlier, in stochastic optics. What is new here is the conjunction: the four fairness theorems and their premises; the commit-rate-independence and kinematic-linearity results that extend exactness to all registration speeds and all quantum sectors; the port-decomposition theorem governing what deviations could ever be observed; the resync-fidelity law $S = 2\sqrt{2}\eta$ for entangled pairs; and a falsifiable deviation package with a tabletop discriminator.

The claim is deliberately bounded. Everything is conditional on a wave-realism premise (P0, §2) and six further premises: four are standard detector physics plus one structural assumption already implicit in perturbation theory (P1–P4; these with P0 carry the entire single-detector theory), and two are ontological commitments consumed only by the multi-quantum and entangled-pair constructions (P5–P6, stated explicitly so that nothing is smuggled). We do not derive probability from a probability-free theory — no one can; we do not derive quantum mechanics; we do not derive entanglement. What we show is that, *given the premises, the measurement postulate is redundant*: definite outcomes, Born weights, Glauber counting statistics, and Bell correlations follow from dynamics plus noise — and, unlike a postulate, the mechanism specifies the conditions under which its statistics would fail, concentrated in one experimental family and testable on a benchtop (§8).

The paper proceeds as follows. §2 states the premises. §3 and §4 develop capture and the selection game, including the instructive failures that make fairness a knife-edge. §5 proves the four fairness theorems. §6 verifies the timescale window in a real detector and extends the mechanism to multi-quantum states, recovering Glauber statistics and ρ -linearity. §7 treats entangled pairs: the two-stage joint game, the fidelity law, the resync mechanism, and no-signaling as a theorem. §8 assembles the deviation ledger, the protective port-decomposition, the tabletop discriminator, and the fork the mechanism cannot yet resolve. §9 situates the result and lists open problems.

2. Premises

Our claim has the form of a conditional: *given four physical premises about detectors, the measurement postulate is redundant* — definite outcomes, Born weights, photodetection counting statistics, and Bell correlations follow from dynamics plus noise. This section states the premises. Each is a statement about the physics of detection; none mentions outcome probabilities, and the quantity $|A|^2$ is nowhere given a probabilistic role by hand. The premises are the interface: everything that follows can be evaluated as a conditional result on P1–P6 alone, independently of any particular ontology supplying them (one such program is noted in §9).

Throughout, a measurement event is decomposed into three stages — **capture** (the incident wave reversibly excites the detector), **selection** (one site ends up with the quantum), and **registration** (the outcome becomes irreversible and macroscopic). Irreversibility enters only at the last stage; that capture is reversible is an experimental fact, not a modeling choice (§3). The premises attach to specific stages; the theorems of §5 govern the selection stage, which is where the Born weights are set.

Notation. An incident single quantum of frequency ω (energy $E_0 = \hbar\omega$) illuminates a detector containing N candidate absorber sites (bound electrons or atoms), indexed i . The local wave amplitude at site i is A_i . During selection, site i holds excitation energy $e_i(t) \geq 0$; we write $E = \sum_j e_j$ for the total and $s_i = e_i/E$ for the *shares*. Outcome probabilities are denoted P_i .

P0 (Wave realism). *The incident quantum is a real, spatially distributed field configuration — not a probability amplitude over mutually exclusive absorbers. Consequently the capture energies $e_i(0)$ are simultaneously, physically present at every illuminated site.* We state this explicitly because it is the ontological commitment on which the identification of an energy-share with an outcome-probability rests. It is not standard-textbook quantum mechanics (which holds the pre-registration state indefinite); it is the realist premise of the substrate program of §9.3, and every result below is conditional on it. P0 is what distinguishes the present mechanism from a mere re-labeling of $|A|^2$: given P0, the $e_i(0)$ are real stakes and the selection dynamics is a genuine physical process, not a bookkeeping identity.

P1 (Energetics of capture). *The incident wave couples to every site within its bandwidth, driving each as a damped oscillator; the energy deposited at site i during capture is proportional to the squared local amplitude,*

$$e_i(0) \propto |A_i|^2.$$

This is elementary driven-oscillator energetics — the energy of an oscillator is quadratic in its displacement — and it is where the squaring enters the theory: as mechanics, not as a probability rule. The same fact appears as the small-tip-angle excitation law in magnetic resonance ($\propto \sin^2(\theta/2) \approx \theta^2/4$) and as the semiclassical photodetection rate $\propto |E(\mathbf{r})|^2$ of Lamb and Scully [Lamb & Scully 1969] and Mandel–Sudarshan–Wolf [Mandel et al. 1964]. We emphasize a discipline that P1 imposes: at the capture stage *every* illuminated site is excited, in proportion to the local intensity. The single-site character of the eventual event is the *output* of selection, never an input. (Reducing the perturbative sum over sites to a single absorption diagram already presupposes the Born rule; we do not do so.)

P2 (Amplitude-linear coupling). *All couplings in the selection dynamics — the drive, site–field exchange, and the vacuum/environment noise — act linearly on wave amplitudes.* Consequently a noise force $\xi(t)$ does work on site i at a rate proportional to its amplitude $\sqrt{e_i}$, so the stochastic increment of the site energy has the form

$$de_i = (\text{drift terms}) + \sigma \sqrt{e_i} dW_i,$$

with energy-noise *variance* proportional to e_i . P2 is not introduced for the present purpose: it is the same structural assumption that underlies golden-rule transition rates throughout perturbation theory — couplings act on amplitudes, and rates emerge quadratic. The burden of this paper is to show that this familiar assumption *also* fixes the outcome probabilities; the sensitivity of that result to P2 — and the failure of every alternative noise scaling — is quantified in Theorem 1. (A physical origin for exactly this scaling — the interference cross term between a site’s complex amplitude and the uncontrolled-phase fields at the surface — is developed in §4.)

P3 (Detector initial conditions). (a) *The internal phases of the absorber sites are initially mutually incoherent: no phase relationship exists among sites, or between any site and the incident wave, before capture.* (b) *The noise driving the selection dynamics is dominantly local: writing the site-noise correlation matrix as C_{ij} , the spatially correlated (radiative) fraction of the noise power is small, $f = \Gamma_{\text{rad}}/\Gamma_{\text{tot}} \ll 1$, so that $C \approx \mathbb{1}$ at working accuracy.* Both clauses are checkable statements about real detectors. (a) holds for any absorber ensemble in internal equilibrium — mutual coherence among detector electrons is not a natural state and would have to be engineered. (b) holds because the linewidths of solid-state absorbers are dominated by local scattering (phonon, carrier–carrier; correlation lengths of nanometres; $\Gamma_{\text{loc}} \sim 10^{13}–10^{14} \text{ s}^{-1}$) while the wavelength-scale-correlated radiative channel contributes only $\Gamma_{\text{rad}} \sim 10^8 \text{ s}^{-1}$: $f \sim 10^{-6}$. Theorem 3 computes the exact penalty when either clause fails — P3 is where the mechanism is *falsifiable*, and §8 [to draft] builds its experimental proposals from precisely these failure modes.

P4 (Registration taxonomy). *Registration is of one of two kinds, according to the absorber’s spectrum.* (i) *Discrete-level absorbers (atoms, molecules): no stationary state exists below the transition energy, so registration requires the full quantum to be assembled at one site.* (ii) *Continuum absorbers (semiconductors, metals): final states exist at all energies above the gap, and registration is a rate process operating throughout selection.* P4 asserts only the taxonomy. The rate law of class (ii) — that commitment occurs at a rate linear in the site occupation — is **not** assumed; it is derived from P2 and energy conservation in Theorem 5, and the outcome statistics are shown to be independent of the rate’s overall magnitude in Theorem 4.

The final two premises are of a different character: they are the ontological commitments consumed only by the multi-quantum and entangled-pair constructions (§6.2–6.3 and §7), and they are stated here so that the interface is complete and nothing is smuggled in later.

P5 (Shared registry). *For multi-quantum and multi-partite states, the joint amplitude — the pair registry $\psi(i, j)$ of a two-quantum state; the joint coefficients c_{ij} of an entangled pair — is physically instantiated in the medium connecting the subsystems, and a registration event renormalizes it faithfully: the surviving joint amplitude re-forms around the realized outcome, so that the conditional weights delivered to the remaining subsystems are exact (update fidelity $\eta = 1$).* P5 is the locus of the mechanism’s nonlocality, held explicitly

rather than implicitly; §7.2 shows that η is experimentally measured to be $\gtrsim 0.99$, and §7.3 discusses why its value should be exactly one.

P6 (Ordering foliation). *Registration events are ordered by a global foliation (a preferred time-slicing); for spacelike-separated commitments, “first” refers to this ordering.* P6 is consumed only in §7, and only to make the two-stage construction well-defined; §7.1 shows the outcome statistics are independent of the ordering, and §7.3 discusses the experimental status of preferred-frame effects.

Three remarks on the status of the premises. First, they are logically independent handles: each theorem of §5 consumes specific premises, and each experimental proposal of §8 targets a specific premise’s failure. Second, P1–P4 are not exotic: P1 and P4 are standard detector physics, P3 is standard detector *matter* physics, and P2 is the structural assumption already underlying the perturbative rate laws of ordinary quantum optics. Third, P5 and P6 are where the premises exceed standard physics, and they are quarantined: no single-detector result (§§3–5, §6.1) uses either. Any ontology supplying P5–P6 supports the constructions built on them; one concrete realization — a substrate framework in active, publicly documented development — supplies all six premises and motivated this investigation [Framework repository; see §9].

3. Stage 1 — capture

When the incident quantum reaches the detector, its wave extends over the whole illuminated patch, and every candidate site within its bandwidth responds. Each site is a bound charge with a transition frequency near ω ; the wave drives each as a damped oscillator, inducing a polarization of amplitude proportional to the *local* field amplitude A_i , with the wave’s phase imprinted on the response. The situation is precisely that of an RF pulse in magnetic resonance: the pulse tips every spin in the excited slice, each by an angle set by the local B_1 amplitude, and stamps its own phase on the resulting precession. At the end of capture the detector carries a *distributed* polarization pattern mirroring the incident intensity profile. No site has been singled out.

We flag a discipline that this picture imposes and that the paper maintains throughout. It is conventional to speak of “the photon interacting with an electron,” as in the one-photon–one-vertex absorption diagram. But that diagram is one term in a coherent sum over all sites, and the step that reduces the sum to a single term — assigning the event to *this* electron with weight $|A_i|^2$ — is exactly the Born-weighted selection whose origin is in question. To assume it at capture is to beg the question the paper answers. The plural must be kept alive until the dynamics kills it; selection is the *output* of Stage 2, never a kinematic input.

The energy law (P1). For a resonant secular drive, the induced oscillation amplitude at site i is proportional to A_i , and the stored energy is quadratic in the amplitude:

$$e_i(0) = \kappa |A_i|^2,$$

with κ common to all sites (and hence cancelling from the shares s_i). The relative phase between the drive and a site’s internal clock affects only the transient, not the secular tip: the drive imprints its own phase, so the deposited energy is phase-independent — a point that matters in §5.2, where the *absence* of surviving phase relationships does real work. In magnetic-resonance language, a small tip angle $\theta \propto B_1 t$ excites with probability $\sin^2(\theta/2) \approx \theta^2/4$ — quadratic in the drive. This is the only place the squared amplitude enters the theory, and it enters as oscillator mechanics. Holography is the same fact as an engineering discipline: recording media respond to the local squared field amplitude — and the fringes survive the exposure’s time integration only because reference and object share a fixed phase, mutual coherence being what renders the energy distribution stationary and hence recordable [Mandel & Wolf 1995]. Computer-generated and digital holograms, in which the pattern is computed from wave theory, fabricated, and successfully reconstructed, verify the quadratic law end-to-end in a setting with no probabilistic reading. (At single-photon exposures, which grain records each quantum is the selection problem of §4; and coherence here is between the fields at capture — pre-established coherence among the absorbers themselves is the separate, testable matter of P3(a) and §8.) Holography also supplies the *control case* for a distinction developed in §4: the single coherent reference is what makes the recorded interference pattern *invertible* — the stored fringes point back to the wavefront, and reconstruction (including phase-conjugate readout, which runs the wavefront backward) is possible precisely because every grain’s record refers to one shared phase standard. An ordinary detector

bulk is, by P3(a), the opposite configuration — each site beats the incident wave against its own uncontrolled reference — and §4 traces what becomes of the phase record in that case.

Relation to semiclassical detection theory. Capture, so described, is the semiclassical photodetector of Lamb and Scully: a quantized absorber driven by a (classical) field reproduces the photoelectric threshold, prompt emission, and a response proportional to $|E(\mathbf{r})|^2$, with no photon concept required. We adopt that result and its credit wholesale. What semiclassical theory famously lacks is *exclusivity*: a classical pulse carrying one quantum of energy, split between two separated detectors, excites both and forbids neither from registering — whereas experimentally a single photon never produces a coincidence (anticorrelation at a beamsplitter). Supplying exclusivity — one full quantum at one site, the rest relaxed — *without importing the Born rule to do it* is precisely the job assigned to Stage 2. Note finally that energy conservation fixes the total: the captured energy sums to the one quantum, $E(0) = \hbar\omega$, which is why the normalization $\sum_i P_i = 1$ will come out automatically rather than by decree.

Capture is reversible. Nothing in Stage 1 is a commitment. The distributed polarization is a coherent excitation of ordinary wave mechanics, and it can be run backwards — not merely in principle but on the bench. The spin echo [Hahn 1950] is precisely this demonstration in the magnetic-resonance setting: excitation deposited across an ensemble, apparently lost to dephasing, is refocused and *re-emitted*, no registration ever having occurred. Its optical twin, the photon echo [Kurnit et al. 1964], shows the same reversal for optical absorbers, and atomic-frequency-comb quantum memories complete the point at the single-quantum level: a single photon is captured across a large ensemble of absorbers and later re-emitted on demand, with the storage step deliberately engineered to be selection-free [de Riedmatten et al. 2008; Afzelius et al. 2009]. Capture-without-registration is thus a working technology, and it fixes an important interpretive point: the deposition of energy across the detector face is *not* the collapse, and the capture stage sits entirely on the settled side of §1’s partition — semiclassical in its response law and reversible in its dynamics. Irreversibility, and with it the arrow that makes an outcome an outcome, enters only at registration, at the bottom of the temporal ladder of §6.1. Everything contested in this paper happens in between.

4. Stage 2 — selection as a stochastic energy game

Capture leaves the detector with tip energies $e_i(0) \propto |A_i|^2$ summing to one quantum; observation finds *one* site registered with the full quantum and the rest relaxed. Stage 2 is the dynamics connecting the two, and its physical content is **exchange**: the tipped sites are coupled to the common field and, through it, to one another; vacuum and environment noise (P2, P3) drives a stochastic redistribution of the deposited energy among them. The process terminates at one of the absorbing configurations supplied by P4: a site that assembles the full quantum crosses the locking threshold and registers (class (i)), or registration fires as a rate process during the exchange (class (ii)); a site whose energy relaxes to zero drops out of the competition.

Two features of the noise deserve immediate comment. First, its role is that of a *tie-breaker*: an infinitesimal symmetry-breaking seed determines which absorbing configuration a given run reaches, but — as the theorems below make precise — the noise *magnitude* drops out of the outcome statistics entirely. The noise decides *that* symmetric alternatives get broken, not *toward whom*, and it carries no outcome information of its own; it is not a hidden variable, and the mechanism is not a local-hidden-variable model. Second, a guardrail we state now and honor in §7: in spatially extended (entangled-pair) configurations, the relevant fluctuations belong to the *shared* substrate, not to independent local reservoirs at each station — a locality caveat that Bell’s theorem makes mandatory for any mechanism of this kind.

The physical origin of the noise: interference at the surface. The interaction at the detector surface is a complex-amplitude affair, and keeping it complex is what identifies the noise physically. Write the complex oscillation amplitude of site i as a_i , so $e_i = |a_i|^2$, and let the site sit not only in the incident wave but in everything else present at the surface: the wave packets of the detector’s other bound charges and the vacuum/environment field, contributing over a short interval a fluctuating complex amplitude increment δb_i of unknown magnitude and — by P3(a) — uncontrolled phase. The site’s energy update then contains

$$|a_i + \delta b_i|^2 = |a_i|^2 + 2 \operatorname{Re}(a_i^* \delta b_i) + O(|\delta b_i|^2),$$

and the interference cross term is the noise: it has *zero mean* because the surface holds no phase reference

against the site (P3(a) — averaging over the phase of δb_i kills it in expectation), and its fluctuation is linear in $|a_i| = \sqrt{e_i}$, so the energy-noise *variance is proportional to e_i* — precisely the P2 form whose uniqueness Theorem 1 establishes. On this reading P1 and P2 are the two halves of one complex expression: the direct term $|a_i|^2$ is the capture law, and the cross term with the uncontrolled-phase surroundings is the fair noise. Note what the argument does *not* require: the magnitudes, spectra, or detailed wave packets of the surrounding charges and vacuum modes never enter — the noise magnitude drops out of the outcome statistics entirely — only linearity (P2) and phase-randomness (P3(a)) are consumed, and both are premises already in place. The unknowability of the surface fields is thus load-bearing rather than regrettable: the martingale theorem makes every property of the noise beyond its first two moments provably irrelevant, so fairness is a *universality* statement — the odds depend only on a symmetry (no phase reference at the surface), the way diffusion depends on molecular chaos and not on molecular trajectories — and the one dial engineering can turn, the fluctuation amplitude, sets rates, never odds. The fairness of the game is thus the statement that *nothing at the surface holds a phase reference against the players*; correspondingly, the engineered violation of P3(a) is a known instrument, not a pathology — beat the signal against a strong coherent reference wave (a *local oscillator*, the homodyne detector of quantum optics) and the same cross term stops averaging away and becomes a signal *linear* in the complex amplitude, which is exactly how homodyne detection behaves [Mandel & Wolf 1995]. One complex expression, two limits: uncontrolled surface phases give intensity capture plus fair noise; an engineered phase reference gives linear field readout. We present this as a mechanism sketch — a scaling argument grounding P2’s form, not the closed microscopic derivation with conservation and drift terms tracked, which remains open (§9.4); the panel’s identification of that derivation as the central open problem stands.

The companion terms and the fairness window. The same expansion that produces the cross term produces two companions, and the sketch is honest only if it tracks them. The bath’s own term $|\delta b_i|^2$ has a mean — heating, and with it the dark counts of a warm detector, which are the selection game won by the bath with no incident quantum — and a fluctuating part whose variance is *independent* of e_i : an additive noise floor. Theorem 1 computes exactly what an additive component does to the game: it drifts shares toward the leaders. Fluctuation–dissipation (the theorem locking the strength of the drag a system feels to the strength of the jiggle it receives) likewise obliges the damping $-2\gamma e_i dt$ to accompany the noise; being linear in e_i , it is share-neutral for site-uniform γ (Appendix B.3). The mechanism’s own logic therefore imposes an operating condition: Born statistics are clean where the stake dwarfs anything the bath can play with — site energies far above the bath’s own scale, which for a thermal environment is the condition $\hbar\omega \gg k_B T$. This is not an embarrassment but a consistency check against the entire detector industry: at 300 K, $\hbar\omega/k_B T \approx 96$ for a 500-nm photon, and optical photon counters work at room temperature; ≈ 5 at 10 μm , and mid-infrared detectors must be cooled; $\ll 1$ at microwave frequencies, and microwave photon counting requires millikelvin cryostats. The window in which the game is provably fair is the window in which quantum detectors are, in fact, operated — and detector cooling, on this reading, protects the statistics themselves, not merely the dark-count rate. The boundary of the window is a candidate signature (§8.6(v)); the closed calculation tracking all three terms in a conserving, quantized model is the open problem of §9.4.

History bookkeeping: the diffused hologram. [Added in v0.5.3; sponsor-originated interpretation, labeled as such — an organizing reading of the mechanics above, not an additional derivation.] The cross-term picture assigns a fate to the incident wave’s *phase history*. In a hologram (§3), one shared reference makes every grain’s interference record mutually registered: the ensemble record is invertible, the stored history points back to the wavefront, and capture can be run in reverse. In a detector bulk satisfying P3(a), each site — and each subsequent hop of the exchange — beats the wave against its *own* uncontrolled reference: every site writes a perfectly good local interferogram relative to a phase standard nobody recorded. Such a *diffused hologram* cannot be inverted; no operation can reassemble the wavefront from it. Three consequences. (i) The same premise does double duty: P3(a)’s reference incoherence is at once why the noise is unbiased (the fair game) and why the record is non-invertible (commitment is possible) — fairness and irreversibility have a common origin. (ii) The phase information is not destroyed but *converted*: the cross terms that would have been recordable fringes under a common reference are exactly the zero-mean fluctuations that drive the selection game — the wave’s own diffused history is the tie-breaker. (iii) The slaved phase of Theorem 2 is, in this language, a *memory*: a sub-threshold site’s phase points back to the drive, which is why κ_{ret} -return is possible at all; an autonomous phase has forgotten the drive, which is why only above-threshold sites

can commit. Reversibility below the layer and commitment above it are the same fact about memory, seen from both sides. The echo and quantum-memory systems of §3 are then *engineered history preservation* — a common or deterministically known reference structure imposed on purpose — and the reversible/irreversible dephasing split of magnetic resonance is the same trichotomy: common reference (invertible), known-different references (invertible by rephasing), random references (non-invertible, the P3(a) case).

The engine that converts this setup into probabilities is classical, and it is not ours:

Theorem 0 (Engine; Pearle 1976, 1982; Adler–Brody–Brun–Hughston 2001). *Let the shares $s_i(t)$ evolve as a martingale (zero drift: $\mathbb{E}[ds_i | \mathcal{F}_t] = 0$) with absorbing configurations at $s_i = 1$ (site i registered) and suppose selection terminates with probability one. Then*

$$P_i = s_i(0) = \frac{e_i(0)}{E(0)} \stackrel{\text{P1}}{=} \frac{|A_i|^2}{\sum_j |A_j|^2},$$

exactly, for any N and any initial distribution. The proof is the optional stopping theorem: s_i is a bounded martingale, so its expectation at the (almost-surely finite) stopping time equals its initial value; at stopping, $s_i \in \{0, 1\}$, so that expectation is P_i . In gambling language: a fair player’s expected fortune is constant, so the probability of eventually holding the whole pot equals the fraction of the chips held at the start. This engine is entirely due to the dynamical-reduction literature — Pearle’s gambler’s-ruin argument and its mature martingale formulation — and we claim no part of it. What that literature postulates, and what §5 derives, is the *fairness*: the zero-drift property, together with its protection against the physical effects that would spoil it.

Fairness is a knife-edge. That the fairness cannot simply be waved through is best shown by exhibiting how decisively natural-looking alternatives fail (simulation details Appendix D). *(a) One-shot selection.* Suppose the winner is simply the site with the largest instantaneous drive overlap, $\arg \max_i A_i |\cos \theta_i|$, with θ_i the random site phases. For two sites this reproduces Born deceptively well (0.791 versus 0.800 for amplitude ratio 2:1 — an accident of arcsine statistics). But on the ten-site test configuration the bright site wins with probability 0.79 against a Born weight of 0.50: extreme-value selection over many draws is dominated by the largest scale, the same disease that makes laser mode competition nearly deterministic. Outcome selection cannot be a snapshot comparison. *(b) Drift-contaminated exchange.* Suppose instead a prolonged exchange of fixed-size energy parcels, clipped so no site goes negative. The clipping makes the zero boundary reflecting — a depleted site can only gain — which is an upward drift for small stakes; the simulated outcome distribution on a three-site configuration is (0.46, 0.27, 0.26) against Born’s (0.64, 0.29, 0.07). A rule that *looks* symmetric fails completely if the martingale property fails microscopically. A stakes-scaled exchange ($\delta \propto \pm \min(e_i, e_j)$, symmetric sign) — the discretization that does satisfy zero drift — reproduces Born within the discretization error ($\sim 2\%$ at coarse step, exact in the continuum limit by Theorem 0).

These failures define the burden of §5 exactly: the physics must place the exchange precisely on the fair point (Theorem 1), protect it against self-amplifying leaders (Theorem 2) and against noise correlations among sites (Theorem 3), and terminate it without distortion (Theorems 4–5). None of these may be assumed; all must come from the premises.

5. The fairness theorems

Every dynamical account of outcome selection before this one obtains Born statistics by *engineering* the noise so that the engine theorem applies; Gisin’s no-signaling results [Gisin 1989; Gisin 1990] explain why they had no choice. The content of this paper is that no engineering is needed: the fairness is forced by the premises of §2. We prove this as four theorems, each answering one physical objection to fairness. The objections are not strawmen — §4’s simulations show that plausible alternatives (extreme-value selection; state-independent noise; correlated noise; nonlinear registration) each destroy Born statistics decisively. Fairness is a knife-edge, and the physics must place the system exactly on it. It does.

5.1 Theorem 1: the noise cannot cheat (fairness from P2)

Theorem 1 (Fair noise). Under P2 and P3(b) (independent site noise), with $de_i = \sigma\sqrt{e_i}dW_i$ and independent Wiener increments dW_i , every share $s_i = e_i/E$ is a martingale:

$$\mathbb{E}[ds_i] = 0 \quad \text{identically, for all } N \text{ and all configurations.}$$

Proof. Itô's lemma applied to $s_i = e_i/\sum_j e_j$ gives $\mathbb{E}[ds_i] = \frac{1}{2}\sum_j \sigma^2 e_j \partial^2 s_i / \partial e_j^2 dt$. With $\partial^2 s_i / \partial e_j^2 = -2\delta_{ij}/E^2 + 2e_i/E^3$,

$$\sum_j e_j \frac{\partial^2 s_i}{\partial e_j^2} = -\frac{2e_i}{E^2} + \frac{2e_i E}{E^3} = 0. \quad \blacksquare$$

The force of the theorem lies in its converse: the $\sqrt{e}$ scaling is the *unique* fair point. For state-independent (additive) noise, the same computation yields $\mathbb{E}[ds_i] = (\sigma^2/E^2)(Ns_i - 1)dt$ — a drift *toward the leaders* (rich-get-richer); for multiplicative noise (variance $\propto e_i^2$), $\mathbb{E}[ds_i] = (\sigma^2 e_i/E^2)(\frac{1}{E}\sum_j e_j^2 - e_i)dt$ — a drift *toward equality*. Both destroy Born statistics, in opposite directions, and simulation confirms the sign and magnitude of each failure: on a ten-site test configuration (nine sites of unit amplitude, one of triple amplitude; Born weight of the bright site 0.500), the $\sqrt{e}$ law reproduces Born at the absorbing boundary (to the Monte-Carlo floor), additive noise drives the bright site toward certainty, and multiplicative noise crushes it to 0.32 (Fig. 2; simulation details Appendix D). (The corrected suite of Appendix D also shows that stopping at a *finite* share rather than at the true absorbing boundary imparts a computable $O(1 - \text{threshold})$ bias — e.g. +2.6% on the two-site 2:1 configuration at threshold 0.95 — which vanishes as the boundary $\rightarrow 1$; the physical boundary is unity because synchronization completes the lock, §6.1.) Amplitude-linear coupling is thus precisely the knife-edge — and it was in the theory before this problem was posed, doing the independent work of generating golden-rule rates. One postulate, both jobs; we regard this coincidence of roles as the strongest internal evidence for the mechanism.

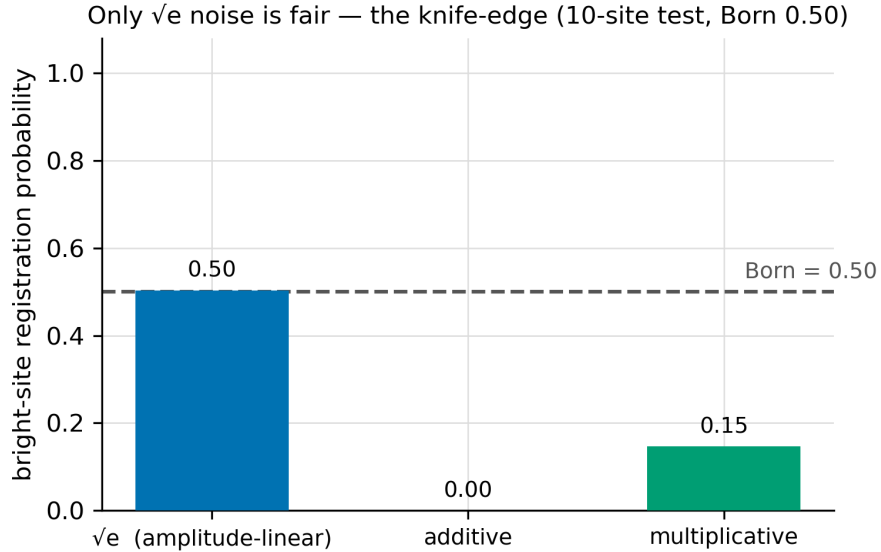


Figure 1: **Figure 2.** The knife-edge (10-site test configuration, Born weight of the bright site 0.50). Only amplitude-linear ($\sqrt{e}$) noise is fair; state-independent (additive) noise drives the bright site toward certainty, and multiplicative noise over-equalizes. Corrected absorbing-boundary simulation.

5.2 Theorem 2: the rich cannot cheat (no autonomous phase below threshold)

The generic disease of winner-take-all dynamics is positive feedback: a leader that couples more strongly *because* it is leading wins with probability approaching one, not with its Born weight (laser mode competition

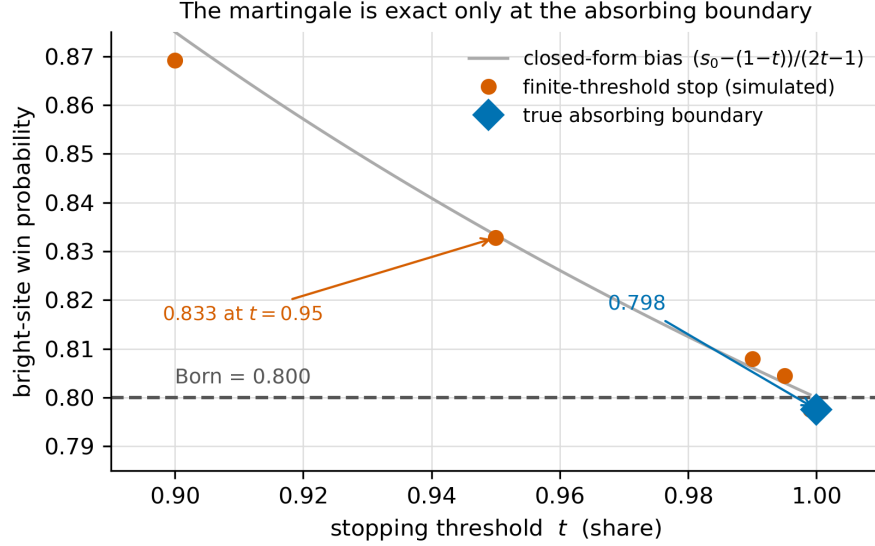


Figure 2: **Figure 3.** The martingale reproduces Born only at the true absorbing boundary. Stopping the game at a finite share and taking the argmax biases the winner probability upward by the closed-form amount $(s_0 - (1 - t))/(2t - 1)$ (grey curve); the two-site 2:1 configuration gives 0.833 at threshold $t = 0.95$, converging to Born (0.80) as $t \rightarrow 1$, and the true absorbing boundary (last survivor) reproduces Born to the Monte-Carlo floor. This is the artifact identified in review and its resolution.

is the canonical example). The bias operates through phase: to extract energy from the common field systematically, a site must *hold a phase relationship* with it. Theorem 2 shows that a sub-threshold site cannot.

Theorem 2 (Slaved phase). *Under P1 and P4(i), a site holding energy $e < E_0$ is off-shell by $\Delta E = E_0 - e$ and constitutes a driven linear mode that returns its excitation to the common field at the rate $\kappa_{\text{ret}} = \Delta E/\hbar$. In the rotating frame, with amplitude $a = re^{i\varphi}$, detuning Δ , and drive $fe^{i\varphi_d}$:*

$$\dot{r} = -\kappa_{\text{ret}} r + f \cos(\varphi_d - \varphi), \quad \dot{\varphi} = \Delta + \frac{f}{r} \sin(\varphi_d - \varphi).$$

In Cartesian coordinates this system is linear with eigenvalues $\lambda = -\kappa_{\text{ret}} \pm i\Delta$; it possesses a unique, globally attracting fixed point $r^ = f/\sqrt{\kappa_{\text{ret}}^2 + \Delta^2}$, $\varphi^* = \varphi_d - \arctan(\Delta/\kappa_{\text{ret}})$. The phase is an enslaved coordinate: perturbations decay at rate κ_{ret} , and no winding (phase-slip) solutions exist.*

What kind of rate κ_{ret} is — and is not. [Added in v0.5.3, prompted by finding N2 of the companion measurement-structure paper’s review panel; clarification articulated by the sponsor.] κ_{ret} is *not* dissipation into a bath, and the earlier drafts’ words “damped” and “decay rate” (with the bath-conventional symbol γ) invited exactly that misreading. A sub-threshold site has no stationary state at its partial energy (P4(i)); its excitation is virtual, and $\kappa_{\text{ret}} = \Delta E/\hbar$ is the rate at which the borrowed energy *reversibly returns to the common field* — the beat rate of a failed lock. The returned energy does not leave the system: it re-enters the exchange pool that the selection game redistributes, which is precisely why the shares can remain a martingale (energy that left irreversibly could not be a fair stake). Dictionary, to be maintained throughout: $\kappa_{\text{ret}} = \Delta E/\hbar$, the deficit-induced reversible return rate (state-dependent, vanishing at the shell); Δ , the drive-transition frequency mismatch (an independent parameter — two different mismatches, two symbols); Γ and γ , reserved for genuine dissipative rates (linewidths, bath damping, the $-2\gamma e$ term of §4). We state plainly what remains unproved: that the return rate is $\Delta E/\hbar$, and that reversible give-back may be modeled as a contraction eigenvalue in the site’s reduced description, are *physically motivated ansätze* pending the microscopic derivation now listed in §9.4(viii).

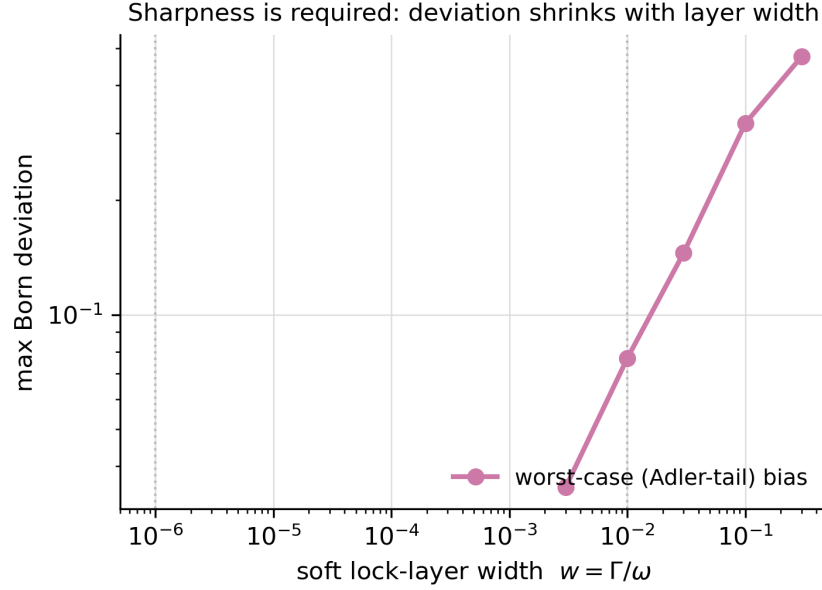
The consequence is structural, not quantitative. The Adler entrainment average $\langle \sin \varphi \rangle \approx K/2\Delta\omega$ — the

standard expression for the systematic pull an oscillator exerts under injection, and the rich-get-richer channel here — is derived from *running* phase solutions of a limit-cycle oscillator, whose phase is a zero mode free to wind. Below threshold there is no limit cycle, no zero mode, and no running solutions: the entrainment tail is, *at linear order in this semiclassical analysis*, not small but absent. Whether a fully quantized common field admits a weak sub-threshold channel — via virtual-photon-mediated interaction — is not settled by the linear analysis; a controlled open-system calculation is required, and is listed among the open problems (§9.4). We therefore claim the tail absent to linear order and bounded by the layer width w beyond it, not absent unconditionally. A free clock — the prerequisite for biased energy extraction — emerges only where $\kappa_{\text{ret}} \rightarrow 0$, i.e. within a boundary layer

$$\Delta E \lesssim \hbar K \quad \Longleftrightarrow \quad w \equiv \frac{\Delta E_{\text{layer}}}{E_0} = \frac{K}{\omega}$$

of the shell, where K is the site–field coupling rate. Here P4(i) does decisive work: a discrete-level absorber has *no stationary state below the transition*, so partial winnings cannot be banked — sub-threshold excitation is virtual, its phase slaved to the common drive (identically for all sites, hence conferring no competitive advantage), its energy decaying on the uncertainty time $\hbar/\Delta E$. Level discreteness enforces threshold sharpness.

Three supporting results complete the argument. *(i) Feedback hierarchy:* from $\dot{e} = 2r\dot{r}$, a slaved site’s uptake from the common field scales as $f\sqrt{e}$ — concave, hence *share-equalizing* (and equalizing toward the drive pattern, which by P1 is the Born pattern — harmless); its return $\kappa_{\text{ret},e}$ is exactly share-neutral; only autonomous (self-sustained) gain, confined to the layer, is share-amplifying. Rich-get-richer requires super-linear uptake, which a linear response cannot supply. *(ii) Quantified tolerance:* simulating the selection game with a rich-get-richer bias confined to a layer of width w gives Born deviations bounded at the percent level even for $w = 0.1$ and at the numerical floor for physical widths ($w = \Gamma/\omega \sim 10^{-6}$ – 10^{-8} for atomic lines, $\sim 10^{-2}$ for broadband solids); by contrast, a bias with the (structurally absent) sub-threshold Adler tail would produce deviations an order of magnitude larger (Fig. 4).



atomic
lines

broadband
solids

(iii) *Continuum absorbers* (P4(ii))

evade the issue entirely: their registration statistics are commit-rate-independent (Theorem 4), so no biased late game exists to be played.

5.3 Theorem 3: the players cannot collude (noise locality)

Theorem 1 requires *independent* site noise. Vacuum fluctuations, however, are correlated over $\sim \lambda/2$; Theorem 3 computes exactly what such correlations do, and what protects real detectors.

Theorem 3 (Correlated-noise drift). *With $de_i = \sigma\sqrt{e_i}dW_i$ and $\langle dW_i dW_j \rangle = C_{ij}dt$,*

$$\mathbb{E}[\dot{s}_i] = \frac{\sigma^2}{E^2} \left[s_i(\sqrt{e} \cdot C \cdot \sqrt{e}) - \sqrt{e_i}(C\sqrt{e})_i \right],$$

which vanishes identically for $C = \mathbb{1}$ and is linear in C . (Proof: Appendix A.) For two sites with correlation c , the drift reduces to $\propto c\sqrt{e_1 e_2}(e_1 - e_2)$: rich-get-richer for $c > 0$, equalizing for $c < 0$.

Correlated noise is genuinely dangerous: with fully correlated noise ($C_{ij} = 1$) the brightest site wins with probability one, and in a resolved fringe geometry at spacing $d = \lambda/2$ a fully-radiative-noise model shows a +20% bright-favoured violation (a continuum of site separations defeats the apparent protection of the $\text{sinc}(k\Delta r)$ zero at $\lambda/2$). What protects real detectors is not geometry but **locality of the dominant noise** (P3(b)): because the drift is linear in C , the deviation for a mixed kernel $C = (1-f)\mathbb{1} + fC_{\text{rad}}$ scales linearly with the correlated fraction f (verified numerically over two decades of f ; Appendix D), and $f \sim 10^{-6}$ for solid-state absorbers puts worst-case Born violations near 10^{-7} . Two further results feed the predictions of §8: the *vector* vacuum kernel is polarization-dependent — at $\Delta r = \lambda/2$ the correlation is $-3/2\pi^2 \approx -0.15$ for dipoles perpendicular to the separation but $+3/\pi^2 \approx +0.30$ parallel, so the deviation’s *sign* rotates with polarization — and absorbers whose noise is *purely radiative* ($f \rightarrow 1$: free-space sub-wavelength atom arrays) are unprotected, making them the natural test bed.

5.4 Theorems 4 and 5: the finish line cannot distort (commitment)

Theorem 4 (Commit-rate independence). *Let registration fire at an arbitrary (possibly time-dependent) total rate, with conditional site-selection probability equal to the share s_i at the moment of firing — the property delivered by any rate law of the form $\text{rate}_i \propto e_i$. Then, with s_i a martingale (Theorem 1) and the commit time T a stopping time,*

$$P_i = \mathbb{E}[s_i(T)] = s_i(0),$$

exactly, at any commit speed. Proof: optional stopping, applied at T rather than at absorption; the conditional pick probability being $s_i(T)$ makes P_i the expectation of a bounded martingale at a stopping time. ■

Theorem 4 reduces the entire registration question to a single one: *is the commit rate linear in the site occupation?* This is not assumed (cf. P4) but derived:

Theorem 5 (Kinematic linearity). *In the n -quantum sector, under P2, a registration channel consuming $k \leq n$ quanta at a site proceeds through exactly k absorption vertices, and its rate is therefore linear in the k -quantum coincidence weight $|\psi^{(k)}|^2$ at that site — never a higher power. In particular, for a single incident quantum, $\text{rate}_i = \Gamma e_i / E_0 [1 + O(w^2)]$.*

Argument. Each vertex is amplitude-linear (P2), so a k -vertex commit amplitude carries exactly one factor of the k -quantum amplitude. A hypothetical rate $\propto e_i^2$ in the one-quantum sector would require a two-absorption-vertex amplitude, which deposits $2\hbar\omega$ into a final state that does not exist: it is blocked by energy conservation *exactly*, not approximately. Absorption–emission combinations return quanta to the pool — they are the exchange dynamics of §4, not registration. The leading correction is a three-vertex (absorb–absorb–emit) path, suppressed by $(\Gamma/\omega)^2 = w^2 \lesssim 10^{-4}$ even for broadband silicon. Saturation corrections require occupied final states and vanish for a ground-state detector. ■

Simulations confirm the theorem pair and its converse (Appendix D): with $\text{rate} \propto s^\alpha$, the case $\alpha = 1$ reproduces Born at every commit speed tested, while $\alpha = 2$ ($\alpha = \frac{1}{2}$) produces bright-favoured (dim-favoured) deviations matching the closed-form fast-commit predictions ($P_1 = 0.941$ predicted, 0.938 observed, at $\alpha = 2$, two sites of Born weights 0.8/0.2) and relaxing toward Born as commitment slows. Both registration classes of P4 thus deliver exact Born by different corollaries of the same martingale property: discrete-level absorbers by first passage (Theorem 0 with absorption at $s = 1$), continuum absorbers by stopped rate-linear commitment (Theorems 4–5).

5.5 Synthesis

The four theorems close the four channels by which a physical selection process could deviate from a fair game — biased dice (Th. 1), a self-amplifying leader (Th. 2), colluding players (Th. 3), and a distorting finish line (Th. 4–5) — and each closure is purchased by an identified physical premise rather than by fiat: P2 buys Theorems 1 and 5, P1+P4 buy Theorem 2, P3 buys Theorem 3. The fairness that dynamical-reduction models must engineer into their noise is here a consequence of detector physics. By the engine theorem, the outcome probabilities are the initial energy shares; by P1, the initial energy shares are $|A_i|^2 / \sum_j |A_j|^2$. That

is the Born rule — with every step either a stated premise or a proved theorem, and with each premise’s failure mode a computable, testable deviation (§8).

6. Reproducing quantum statistics

The theorems of §5 are exact in their stated limits; a physical detector must realize those limits within its actual hierarchy of rates, and the mechanism must extend beyond single quanta to the full counting statistics of quantum optics. This section verifies both.

6.1 The timescale window in a real detector

We take a silicon single-photon avalanche diode at $\lambda = 500$ nm as the working example ($\omega = 3.8 \times 10^{15}$ rad/s; $E_0 = 2.5$ eV). The relevant ladder:

- *Exchange step*: by fluctuation–dissipation, the noise driving the game has variance rate set by the final-state width, $\Gamma \sim 1.5 \times 10^{13}–10^{14} \text{ s}^{-1}$ (widths of 10–100 meV): one exchange step per 10–70 fs.
- *Game duration*: the selection must transport log-shares a distance $\ln N$ diffusively, giving $\tau_{\text{game}} \sim 2 \ln^2 N / \Gamma \approx 12–81$ ps for $N \sim 5 \times 10^{10}$ sites in the diffraction volume, across the stated Γ range (a device-specific estimate, not a substitute for a detector-specific open-system calculation). (Only $\ln^2 N$ enters; the answer is insensitive to the coarse-graining of “sites.”)
- *Losses*: silicon’s indirect gap puts radiative loss at microsecond scales, and multiphonon dissipation of a 2.5 eV *virtual* excitation is blocked by the absence of sub-gap final states — the same discreteness that sharpens the threshold (Th. 2) protects the stakes. Loss fraction during the game: $\lesssim 10^{-5}$.
- *Pattern-degrading transport*: below threshold, coherent transport of the excitation *is* the slaved polariton wave — it carries the Born pattern rather than diffusing it. Incoherent diffusion requires real population, which exists only after commitment: spread $\sqrt{2D\tau_{\text{commit}}} \sim 8$ nm against a micron-scale pattern.

The window thus closes with two to four orders of magnitude to spare at every rung: $1.7 \text{ fs} \ll 10–70 \text{ fs} \ll 12–81 \text{ ps} \ll \text{ns}–\mu\text{s}$. Both orderings of the photon coherence time against the game are benign. A long wavepacket continually re-pins the shares to the instantaneous Born pattern — a re-preparation, not a distortion — and predicts a click-time density proportional to $|\text{envelope}(t)|^2$, i.e. time-resolved Born statistics, as observed. A short pulse leaves the game to run drive-free after capture, the clean martingale regime. (We note without weight that τ_{game} falls in the tens-of-picoseconds range of measured SPAD timing jitter.)

Why the boundary sits at unity. The game terminates when one site holds the whole quantum ($s_i \rightarrow 1$); the simulations (Appendix D) reproduce Born to the Monte-Carlo floor at this absorbing boundary, whereas stopping at any finite share introduces a computable $O(1 - \text{threshold})$ bias. The unit boundary is not a numerical idealization but the temporal face of Theorem 2: below the shell a site has no autonomous phase and cannot sustain a lock, so a partial quantum cannot be banked, and the irreversible lock engages only as a site assembles essentially the full quantum. The residual gap $1 - \theta_{\text{lock}} = O(w)$, with $w = K/\omega$ the boundary-layer width, is then the physical origin of the predicted deviations — negligible for atomic lines ($w \sim 10^{-6}$), percent-level for broadband solids ($w \sim 10^{-2}$). The oversized threshold of a crude stopping rule and the physical deviation are the same expression evaluated at different gaps.

The temporal structure of an event. The ordering of the ladder is not incidental. Capture is bounded below only by the energy–time uncertainty, $\tau_{\text{capture}} \gtrsim \hbar/\Delta E$; for a massless incident quantum, which carries no rest-mass Compton clock, this bound is saturated at $\tau \sim 1/\omega$ — the fastest an interaction can proceed — so the squared-amplitude pattern is imprinted essentially instantaneously and faithfully, before any slower dynamics can bias it. The lock then engages *reversibly* on the fast synchronization scale; irreversibility is supplied only by the slow dissipative commit at the bottom of the ladder, where the assembled quantum is delivered to the detector’s bath and the record is written. Two consequences follow. First, by the commit-rate independence of Theorem 4, the rate of this slow step drops out of the outcome weights: dissipation sets *when* an outcome finalizes, never *toward whom*. Second, the reduced site-energy fails to be conserved (§5.1) precisely because energy leaves the site subsystem into the bath at this step — which locates the conserving

field+detector+bath derivation (§9.4) as one in which the fast capture–selection dynamics is nearly closed and the slow registration is where energy irreversibly flows out. We offer this temporal picture as interpretation consistent with Theorems 2 and 4, not as an independent theorem; the controlled open-system derivation is an open problem (§9.4).

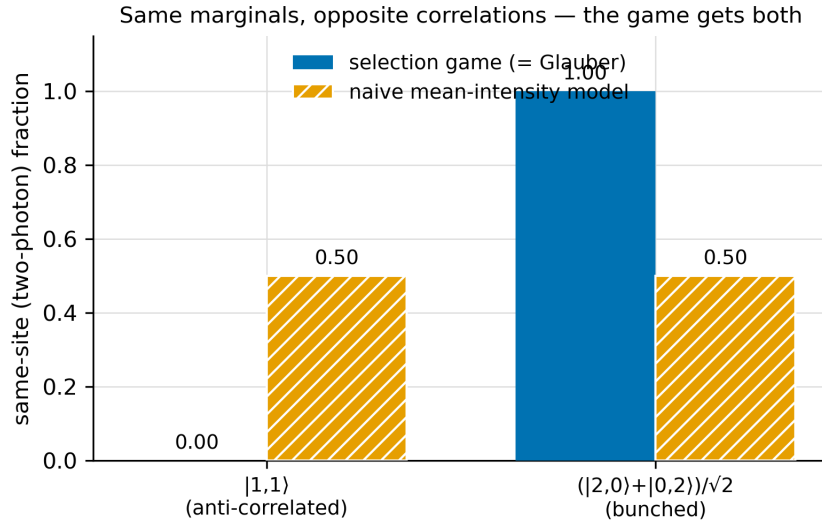
6.2 Multi-quantum states: Glauber counting

For n incident quanta the medium carries the *joint* amplitude (P5) — for two, the symmetric pair registry $\psi(i, j)$ — and detector stakes are drawn from it sector by sector. A one-photon absorber plays for the single-quantum marginal $m_i = \sum_j |\psi(i, j)|^2$. Upon the first commitment, at site i^* , the registry renormalizes (the same conditional update that §7 uses across entangled wings), and the second game runs on the conditional row $|\psi(i^*, j)|^2 / m_{i^*}$. Each game is Born-exact by §5, so the joint distribution is marginal times conditional:

$$P(i, j) = m_i \cdot \frac{|\psi(i, j)|^2}{m_i} = |\psi(i, j)|^2$$

— Glauber’s joint photodetection distribution, from the same composition property that will produce the Bell correlations.

The discriminating test is a pair of states with *identical mean intensities but opposite correlations*, for which any intensity-only model returns identical predictions. For two site-localized modes: the anti-correlated state $|1, 1\rangle$ (one quantum at each site) and the bunched state $(|2, 0\rangle + |0, 2\rangle)/\sqrt{2}$ both give 50/50 marginals. The simulated game yields perfect anti-bunching for the first (the two quanta *never* register at the same site; split fraction 1.000) and perfect bunching for the second (same-site fraction 1.000), reproducing the Glauber predictions exactly, where the intensity-only model predicts an uncorrelated 50/50 split for both (Fig. 5).



For a generic (randomly generated) three-site pair state, the simulated joint distribution matches $|\psi(i, j)|^2$ entrywise to 2×10^{-3} , the statistical floor of the run.

6.3 Nonlinear-intensity detectors and ρ -linearity

A two-photon absorber — a site with no level at $\hbar\omega$ but one at $2\hbar\omega$ — commits through exactly two absorption vertices (Theorem 5), so its amplitude carries exactly one factor of the *pair* amplitude at the site, and its rate is

$$\text{rate}_i \propto |\psi(i, i)|^2 = G^{(2)}(i, i),$$

the standard two-photon POVM: nonlinear in intensity, linear in the pair-sector weight. The registry structure is not optional decoration; it is forced by consistency. On the state $|1, 1\rangle$ the pair diagonal vanishes, $\psi(i, i) = 0$, and the game correctly predicts *no* two-photon registrations at either site — where a model that

squared the mean intensity would falsely predict clicks at both. The simulation confirms both this null and the 50/50 two-photon statistics of the bunched state.

The general statement assembles from Theorem 5 and the convex structure of the ensemble. Every allowed commit channel's rate is linear in its coincidence weight $G^{(k)}$; every $G^{(k)}$ is a linear functional of ρ ; per-configuration outcome statistics are Born (§5); and a mixed state, realized as an ensemble of substrate configurations, averages its predictions affinely (verified numerically to 6×10^{-4}). Hence *all* outcome statistics are affine in ρ — for arbitrary detectors, in every sector — with two immediate consequences: predictions depend on a mixed state only through ρ (decomposition-independence), and the ground is laid for the no-signaling results of §7. The framework thus reproduces the POVM structure of photodetection theory, with the selection mechanism running beneath it rather than instead of it.

7. Entangled pairs and no-signaling

7.1 The joint game

The decisive test of any outcome-selection mechanism is the entangled pair, where Bell's theorem forbids the correlations from arising in independent local games. Consider $|\Phi^+\rangle = (|HH\rangle + |VV\rangle)/\sqrt{2}$ distributed to stations A and B with analyzers at angles a, b , each followed by a two-port detector. Two structural facts dictate the mechanism's shape. First, each wing's *local* wave is unpolarized — its reduced state gives 50/50 shares in every basis — so by §5 the local games can produce only the correct *marginals*, which carry no angle information. This is as it must be: local statistics in a Bell experiment are coin flips. Second, the joint weights $P(\pm\pm) = \frac{1}{2} \cos^2(a-b)$ reside in the *joint* amplitude, which by P5 is physically instantiated: the pair was created as a single excitation of the shared medium, and the two waves retain a stored phase relationship — the same registry structure that carried the pair amplitude $\psi(i, j)$ in §6.2. This shared structure is explicit and real — a correlation that local bookkeeping ignores, not a hidden variable.

The selection process is then two-stage. Both local games run concurrently. Whichever wing *commits first* — the ordering supplied by the foliation of P6 — undergoes a lock that renormalizes the pair registry: the joint amplitude re-forms around the realized outcome, as dynamics rather than decree, and the far wing's shares are thereby reset to the conditional Born weights (e.g. $P(B=+ | A=+) = \cos^2(a-b)$). The far wing's game then completes on the updated shares. Because a conditional reset is simply a new initial condition, and the theorems of §5 hold for *any* initial condition, both factors are exact:

$$P(i, j) = \underbrace{m(i)}_{\text{marginal}} \times \underbrace{P(j|i)}_{\text{conditional}} = |c_{ij}|^2,$$

independently of which wing commits first — and order-independence of the statistics is precisely the no-signaling property. Simulation of the full stochastic process (two concurrent games, stochastic commit order, conditional resync; Appendix D) gives CHSH $S = 2.820$ at the standard angles against the quantum value $2\sqrt{2} = 2.828$; $E(a_1, b_1) = 0.706$ vs 0.707 ; the A marginal flat under change of B 's setting (± 0.007 , consistent with zero); and identical correlations conditioned on commit order (0.717 vs 0.706). We emphasize that reproducing $2\sqrt{2}$ is a *consistency check*, *not a result* — any Born-faithful mechanism inherits it. The content of this section lies in what follows.

7.2 The fidelity law

Degrading the conditional update — far-wing shares reset to $\eta \cdot (\text{conditional}) + (1 - \eta) \cdot (\text{current})$ — yields $E(a, b) = \eta \cos 2(a - b)$ and hence

$$S = 2\sqrt{2}\eta,$$

verified across the sweep ($\eta = 1, 0.9, 1/\sqrt{2}, 0.5 \rightarrow S = 2.820, 2.533, 2.000, 1.421$): the local-realism bound is crossed exactly at $\eta = 1/\sqrt{2}$. The *resync fidelity* η is thereby promoted from interpretive gloss to physical parameter, and existing experiments measure it: photonic CHSH values near 2.8 over 10^2 -km separations imply $\eta \gtrsim 0.99$. Whatever the substrate's update mechanism is, it is near-perfect at terrestrial scales. (§8.7 proposes a *controlled* measurement of η : deliberately injected broadband cross-station noise leaks into the correlators only through registry infidelity, with a computable, angle-independent signature.)

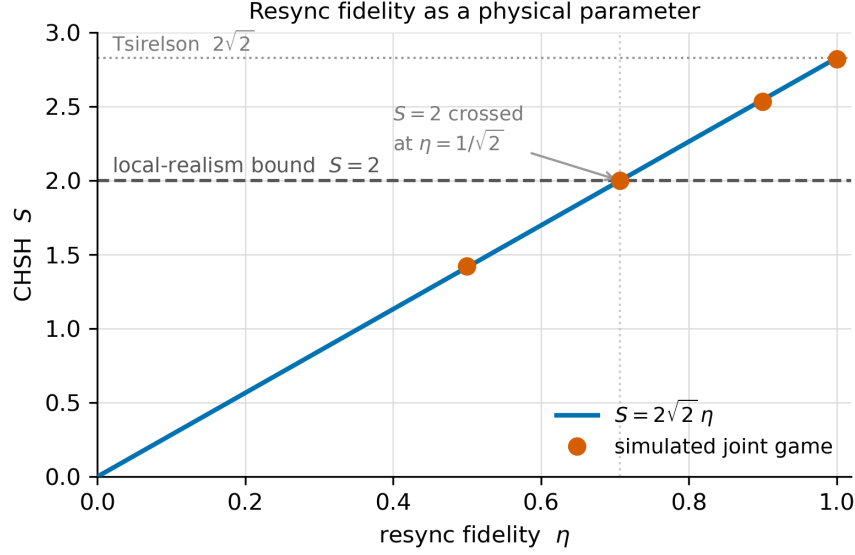


Figure 3: **Figure 6.** The resync-fidelity law $S = 2\sqrt{2}\eta$. Degrading the conditional update to fidelity η scales the CHSH value linearly; the simulated joint game (points) tracks the law, crosses the local-realism bound $S = 2$ exactly at $\eta = 1/\sqrt{2}$, and reaches the Tsirelson value $2\sqrt{2}$ at $\eta = 1$. The fidelity η is thereby a measurable physical parameter, not an interpretive gloss.

7.3 The resync mechanism and the speed bounds

What kind of process could realize the faithful update that P5 asserts? The candidates divide into *propagating* influences (finite $v \gg c$ in the preferred frame) and *global* re-equilibration. The picture adopted here is the latter, in geometric form: the pair is one substrate object whose correlation is a single shared internal orientation — a phase frame anchored at the source event — and the first lock *rotates that shared frame*. The changed degree of freedom is an angle, not a field configuration traversing the intervening space; the apparent superluminality is a category error of the same family as phase-velocity and pattern motions (the intersection point of closing scissors), which move arbitrarily fast precisely because nothing physical transits. Three consequences discriminate this picture: (i) experiments bounding the “speed of quantum influence” can only ever return growing lower bounds — currently $> 5.4 \times 10^4 c$ for a CMB-like preferred frame (Salart et al.) and $\geq 1.38 \times 10^4 c$ (Yin et al.) — and the rotation picture predicts *no floor will ever be found*, whereas any propagating variant must eventually exhibit one; (ii) fidelity does not degrade with distance — rotations do not attenuate — consistent with undiminished violation at satellite scales, and making $\eta = 1$ natural rather than tuned; (iii) “before-before” configurations with moving detectors (Zbinden et al.; Stefanov et al.), in which each wing is first in its own rest frame, are dissolved: the ordering that matters is carried by the foliation at the shared object, not by either detector’s frame — consistent with the null results of those experiments.

7.4 No-signaling as a theorem, and the rigidity of the statistics

The traditional logic (Gisin) runs: no-signaling is imposed, and it forces the linear/martingale structure of any acceptable stochastic dynamics. Here the arrow reverses. Theorem 1 (fair shares) and Theorem 5 (kinematically linear commitment) are *derived* from the premises, they entail statistics affine in ρ (§6.3), and affine local statistics entail that no operation at one wing moves the marginals at the other: **no-signaling follows as a theorem** — but one *conditional on the shared-registry premise P5* (with faithful update $\eta = 1$), which supplies the affine structure in the first place. The logical order is nonlocal ontology (P5/P6) $\rightarrow$ Born-weighted local games $\rightarrow$ affine statistics $\rightarrow$ no-signaling; this is a *consistency* result, not a derivation of no-signaling from local detector physics. It holds exactly at the statistical level over an explicitly nonlocal ontic structure.

The inverse direction quantifies how rigid this is. Introduce a hypothetical commit nonlinearity at wing A , $f(s) = s^{1+\varepsilon}$, and prepare the partially entangled state $\sqrt{p}|00\rangle + \sqrt{1-p}|11\rangle$. The fast-commit limit gives the far-wing marginal in closed form,

$$M(B=0; \alpha) = p \frac{\cos^2 \alpha s_+^\varepsilon + \sin^2 \alpha s_-^\varepsilon}{s_+^{1+\varepsilon} + s_-^{1+\varepsilon}}, \quad s_+ = p \cos^2 \alpha + (1-p) \sin^2 \alpha,$$

which acquires a dependence on A 's setting α — a signal — of magnitude $\Delta M \approx K(p) \varepsilon P_{\text{first}}$ with $K(0.8) \approx 0.18$ and $P_{\text{first}} = \frac{1}{2}$ for symmetric stations (full-game simulation: $\Delta M = +0.044$ at $\varepsilon = 0.5$ against the ceiling $+0.089$; -0.0002 at $\varepsilon = 0$). Two exact zeros structure the phenomenology: $\varepsilon = 0$ (any state — the theorem), and $p = \frac{1}{2}$ (any ε — a symmetry protection at maximal entanglement). The latter has an experimental moral: *maximally entangled Bell tests, which is to say nearly all Bell tests, are the least sensitive configuration for this physics*; the sensitive test is partial entanglement with asymmetric bases, a pointer developed in §8.

7.5 What is derived and what is postulated

Derived: that concurrent local fair games with rate-linear commitment convert the joint amplitudes into exactly Born-weighted, order-independent, no-signaling joint outcomes — the composition property doing for two wings what it did for two quanta in §6.2. Postulated: P5 itself — that the medium carries the joint amplitude between the wings and renormalizes it faithfully ($\eta = 1$) — and the ordering foliation P6. This section is therefore a *consistency proof* of the selection mechanism with the entanglement ontology of P5–P6, not a derivation of entanglement; the nonlocality is not explained away but located — in the shared registry and the preferred foliation — and η , currently an input measured to hold at the percent level, is the natural next target for derivation, with §7.3's rotation picture indicating why its value should be exactly one.

8. Predictions and falsifiability

8.1 The deviation ledger

A mechanism, unlike an axiom, can fail — and the premises of §2 delimit exactly where. Table 1 records every candidate deviation channel this program has generated, including two that the theory itself subsequently eliminated. We display the retractions deliberately: a mechanism should generate candidate failure modes and then be held to its own theorems, and the pattern of closures is itself evidence about the surviving channels.

Table 1 — Born-deviation channels and status.

Channel	Premise at risk	Status
Soft locking threshold (broadband absorbers, layer width $w = \Gamma/\omega \sim 10^{-2}$)	P4	Retracted. Artifact of imposing a hard threshold on a continuum absorber; Theorem 4 makes the statistics commit-rate-independent.
Nonlinear commit rate (cooperative/two-carrier registration), any sector	P4	Retracted. Kinematically blocked by Theorem 5; residual $\varepsilon \lesssim w^2 \lesssim 10^{-4}$.
Correlated vacuum noise (radiative-dominated absorbers, e.g. sub- λ atom arrays)	P3(b)	Live. Deviations $\propto$ correlated noise fraction f ; large only as $f \rightarrow 1$.
Pre-established absorber coherence (phase-locked ensembles)	P3(a)	Live (with the caveat that maintaining coherence requires driving, which exits the single-quantum sector; §8.6).

The surviving channels are exclusively failures of P3 — the Stage-2 fairness conditions. Neither afflicts any detector in experimental use.

8.2 The port decomposition: why the entire experimental record is protected

Deviations of the surviving type bias *which site within a detector patch* registers. But the outcome that experiments record — and that carries correlation information — is the *port* (which analyzer output fired). Summing Theorem 3’s drift over the sites of each port (cross-port noise correlations vanish for macroscopically separated ports) gives, for the port share S_A ,

$$\dot{S}_A \propto q_B S_A - q_A S_B, \quad q_P = \sum_{i,j \in P} C_{ij} \sqrt{e_i e_j},$$

with the immediate consequence: **if the two ports have matched collective structure** ($q_A/S_A = q_B/S_B$ — same geometry, same internal correlation) **the port share is drift-free, and port statistics are exactly Born even when within-port statistics deviate arbitrarily.** Within-port deviations are invisible to every recorded observable and can never signal. All experiments on record use structurally symmetric ports; combined with $f \sim 10^{-6}$ for their detectors, the existing record is protected with roughly six orders of magnitude to spare. The mechanism is not threatened by any measurement yet performed — and, less comfortably, it has not yet been *tested* by one either. The remainder of this section designs the test.

8.3 The live configuration

Let the two ports differ in collective structure — one a dense sub-wavelength array, the other sparse; or one phase-locked to an external reference. Simulation of the port game (Appendix D) gives a bias linear in the mismatch of the within-port correlated noise fraction, $\kappa \approx -\Delta c_{\text{eff}}$, with port statistics

$$P'(S) \approx S + \kappa S(1 - S)$$

(the internally correlated port *suppressed* — a Jensen-inequality effect: correlated noise inflates a port’s energy variance, and shares are concave in energy), together with a state-dependent no-click rate in the strong-correlation limit.

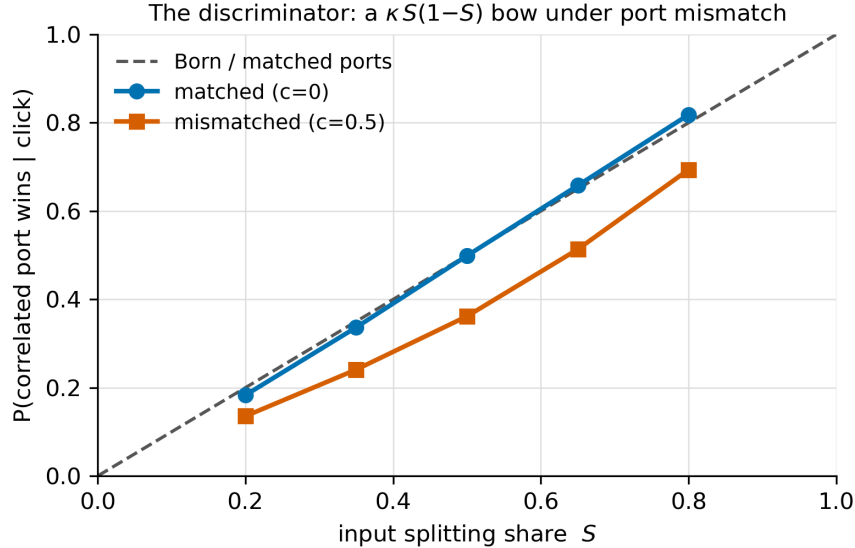


Figure 4: **Figure 7.** The tabletop discriminator. Conditional probability that the correlated port registers, versus the input splitting share S . Matched ports ($c = 0$) track the Born diagonal; deliberately mismatched within-port noise correlation ($c = 0.5$) produces the predicted $\kappa S(1 - S)$ bow, suppressing the internally-correlated port. Any nonzero curvature beyond calibrated efficiency effects discriminates against the linear-POVM structure of standard quantum mechanics.

An essential caution: for radiative-dominated arrays, standard quantum mechanics *itself* predicts collective modifications of absorption — the response is governed by dressed (sub- and superradiant) modes, not independent atoms [Asenjo-Garcia et al. 2017; Javanainen et al. 2014; Javanainen & Ruostekoski 2016] — and the collective response is not hypothetical but measured: a single sub-wavelength monolayer of a few hundred atoms shows subradiant line narrowing and mirror-grade reflectivity [Rui et al. 2020], controllable at the single-photon level [Srakaew et al. 2023]. Deviation from *naive per-site* weighting is therefore ordinary cooperative optics and tests nothing. The framework’s claim is a deviation from the *dressed-mode Born prediction*; the quantitative confrontation of the port game with the collective-mode calculation is open work, and the discriminator below is designed so that this subtlety cannot contaminate it.

8.4 A tabletop discriminator

Send single photons at a variable splitting ratio $S : (1 - S)$ into the two ports of a detector with deliberately mismatched collective structure, the ports spatially separated, and record the conditional port statistics $P'(S)$. For *any* local POVM — i.e., for standard quantum mechanics with arbitrary collective physics inside each port — the cross-port matrix elements vanish and $P'(S)$ is exactly affine in S : a straight line whose slope encodes the (arbitrary, possibly collective) port efficiencies. The framework predicts *curvature*, $\kappa S(1 - S)$, plus a state-dependent no-click anomaly. The signature is thus insensitive to the dressed-mode physics inside each port: any nonzero curvature (beyond calibrated nonlinearities) discriminates against the linear-POVM structure of quantum mechanics itself. No entanglement, no timing, no Bell configuration — a single-photon source, a variable splitter, and counting statistics probe the mechanism’s most radical corner at the $|\kappa|$ level, with candidate implementations (tweezer-array ports at sub- λ spacing, $f \rightarrow 1$) reaching $|\kappa| = O(10^{-1})$ if the live channel is real. The ingredients are separately demonstrated: sub-wavelength atomic arrays with measured cooperative optical response, down to the single-photon level [Rui et al. 2020; Srakaew et al. 2023], and single-shot registration of one absorbed photon by a cold-atom ensemble, at efficiencies from $\sim 20\%$ in free space to $\gtrsim 90\%$ with cavity assistance [Paris-Mandoki et al. 2017; Yang et al. 2022; Vaneecloo et al. 2022]. What remains unbuilt is the combination: an array operated as a *calibrated absorption-counting port*. One distinction matters in assembling it: Rydberg blockade realizes the $f \rightarrow 1$ limit taken all the way to a single effective absorber — erasing the within-port site competition that the port game concerns — so blockaded superatoms supply the readout existence proof, while the deviation medium proper is the un-blockaded array of competing sites.

8.5 The fork

If the live channel is real, its consequences do not stop at Born deviations. A mismatched-port detector at one wing of an entangled pair shifts the far wing’s marginals — and because the asymmetry resides in the *detector*, the maximal-entanglement protection of §7.4 does not apply: $M(\alpha) = \frac{1}{2} + \delta \cos 2\alpha \cdot P_{\text{first}}$, a setting-dependence, i.e. in-principle superluminal signaling. Two readings are open, and we state them as a fork rather than resolve it prematurely. On the *radical reading*, the prediction is genuine; nothing in existing data excludes it, because the configuration has never been built. On the *protective reading*, a deeper fairness principle — for instance, a compensating renormalization of uptake and decay by the same substrate dynamics that correlates the noise — closes this channel as kinematic linearity closed the commitment channel, and the framework becomes *derivably* equivalent to quantum mechanics for all detector configurations, which would itself be the strongest form of the program. Our judgment favors the protective reading: twice already in this work, channels left open by first analysis were closed by theorems (Theorems 4 and 5, retiring the first two ledger entries), and the pattern favors closure. But the framework as here formulated cannot decide, and the experiment of §8.4 can: measured curvature confirms the radical reading (or bounds κ); a null at the 10^{-3} level closes the last deviation channel and renders the mechanism empirically equivalent to standard quantum mechanics in every configuration tested.

8.6 Secondary predictions

For completeness: (i) *time-resolved Born* — click-time density $\propto |\text{envelope}(t)|^2$ (§6.1), already consistent with observation; (ii) *no speed floor* for the “speed of quantum influence” (§7.3) — a standing prediction distinguishing global from propagating resync; (iii) *polarization signature* — where array deviations exist,

their sign rotates with polarization relative to the array axis (the vector vacuum kernels of §5.3: $c_{\perp} \approx -0.15$ vs $c_{\parallel} \approx +0.30$ at $\lambda/2$); (iv) *differential pre-coherence* — for the P3(a) channel, the active variable is the *difference* in pre-established coherence between ports, symmetric pre-coherence being protected by §8.2; the driven (local-oscillator) implementations of pre-coherence add quanta and exit the single-quantum sector, so their analysis requires the multi-quantum machinery of §6.3 and is left open; (v) *warm-detector bias* — **a candidate, not yet a ledger entry**: where site energies approach the bath’s additive noise floor ($\hbar\omega \lesssim k_B T$; §4’s companion-term analysis), the additive component drifts shares toward the leaders (Theorem 1), implying a reproducible odds *bias* toward bright sites over and above the excess dark counts — whereas standard quantum mechanics expects a marginal detector to grow noisier, not biased. Promotion to the ledger requires an analysis, not yet performed, separating this signature from dead-time, saturation, and afterpulsing systematics, and a quantitative treatment of the crossover region where signal stakes and bath floor compete. (vi) *energy-audit interferometry* — **a consistency test, not a discriminator, carrying a sponsor-originated interpretive clause**: send heralded single photons of known energy $\hbar\omega$ through a double slit or interferometer onto an *energy-resolving* detector plane (TES- or MKID-class calorimetry, few-percent resolution or better), and compare each click’s deposited energy against the created photon’s. Standard quantum mechanics and this mechanism predict the same result — the full quantum per click, at every fringe position, never a fractional deposit weighted by the local intensity — so a positive result discriminates *readings*, not theories: it directly excludes any local energy-accumulation picture in which sites keep their $|A_i|^2$ shares, complementing the first-click-latency exclusion of intensity-dependent delay, and it is the energy-domain twin of the spacelike anticorrelation already cited in §3. The mechanism adds one sharper clause — the sponsor’s reading of Theorem 2: the distributed Stage-1/2 excitation is *reversible*, off-shell shares returned to the common field at κ_{ret} , so the energy “in the fringe” is never truly deposited; deposition completes only at registration, and losing sites should retain **no calorimetric residue** (no anomalous heating of dark fringes or of the losing arm) beyond ordinary optical losses. The canonical interference experiments could not ask this question — their click detectors are charge-triggered and energy-blind — and a literature search (2026-07) finds no published experiment framed as this audit; the closest existing evidence is incidental and large: energy-resolving MKID astronomical cameras (ARCONS, DARKNESS, MEC) record per-photon energy, position, and arrival time at focal planes shaped by diffraction, and TES calorimetry routinely confirms $\hbar\omega$ per absorbed heralded photon, while gamma-ray spectroscopy’s photopeak/Compton structure is the same bookkeeping at MeV scale (full quantum at one site; any mismatch departing as real quanta). Promotion beyond consistency-test status requires the residue clause made quantitative — how much transient heating, if any, a calorimeter should register at a losing site during the selection window — which turns on the same κ_{ret} microphysics as §9.4(viii) and is logged with it.

8.7 The cross-station injection test

The proposals of §§8.1–8.4 all target the single-detector premises. The entangled sector rests additionally on P5, and it admits its own instrument: *deliberately couple a common classical perturbation into both detectors of a working Bell experiment* — in the selection band, at the absorbers, bypassing the shielding by design — and read the coincidence statistics. Standard quantum mechanics predicts a null: correlated classical noise at the stations can move *rates* (dark counts, correlated efficiency drift — removable by fast setting-switching and time-tagged analysis) but cannot redirect *outcomes*, so the correlators are untouched. The present mechanism predicts a structured response, computed from the §7 joint game extended with cross-station noise [Framework repository: DERIVATION_two_station_correlated_noise, simulation suite two_station_sims], and the structure separates cleanly into two knobs.

Universal protections. Two results bound what any injection can do. First, each station’s share drift is blind to cross-station covariance, so the *marginals stay exactly Born under any correlated injection whatsoever* — the effect, if any, lives only in coincidences, and the test is invisible in singles. Second, a *protection theorem*: for broadband (white) cross-station correlation of any strength $\rho < 1$, with faithful registry update ($\eta = 1$), the joint statistics remain exactly quantum. The correlation is genuinely generated — in simulation the far wing’s share *leans* toward the near wing’s developing outcome by $L = +0.29$ at $\rho = 0.8$ — and is then discarded, shot by shot, by the conditional reset of §7.1. White noise is additionally memoryless, so non-overlapping selection windows never see it at all. Broadband common-mode pickup between Bell stations is therefore harmless to the outcome statistics *even in-band*: a stronger statement than the engineering

intuition that shielding suffices.

Knob 1 — broadband injection measures the registry fidelity. If the update is imperfect, the reset retains the fraction $(1 - \eta)$ of the far wing’s lean, and the leak is exactly computable: every correlator shifts by the same angle-independent amount, $\delta E = 2(1 - \eta)L(\rho)$, hence

$$\delta S = 4(1 - \eta) L(\rho),$$

confirmed by simulation at the percent level ($\delta S = +0.117 \pm 0.005$ observed against $+0.115$ predicted at $\eta = 0.9$, $\rho = 0.8$, with all four correlators shifted uniformly). Broadband injection at a *known, controlled* ρ therefore converts §7.2’s inference ($\eta \gtrsim 0.99$ from existing CHSH values) into a laboratory dial: any measured δS is a direct reading of $(1 - \eta)$.

Knob 2 — phase-locked injection and the above-Tsirelson signature. A coherent tone, phase-locked between the stations with global phase randomized shot to shot, is not white: its phase persists through the conditional reset and biases the far wing’s *post-reset* game — the one contribution the registry cannot wipe. At leading order each correlator shifts by

$$\delta E(a, b) = 2 \kappa_1 \kappa_2 \lambda^2 \cos 2(a - \psi) \cos 2(b - \psi),$$

with λ the injection’s drift-to-diffusion ratio, ψ its polarization, and $\kappa_{1,2}$ the measured drift susceptibilities of the first- and second-committing wings. Three signatures follow, all verified in the simulated joint game: (i) $\delta S > 0$ — the tone *adds* classical common-bias correlation on top of the quantum correlation, pushing S above the Tsirelson bound $2\sqrt{2}$, which no quantum-mechanical mechanism can do; an $S > 2\sqrt{2}$ reading under injection would be an unambiguous non-QM signature. (ii) The angle-resolved pattern rotates with the injected polarization ψ — a fingerprint tying the deviation to the injection rather than to any systematic. (iii) The singles stay Born throughout (the shot-randomized global phase averages the marginal biases away), so the effect masks itself from every single-station monitor — deviation and no-signaling coexisting in exactly the way the two-stage construction prescribes.

The protocol carries its own calibration. The identical tone applied to *one* station with a *fixed* (not shot-randomized) phase is precisely the engineered-phase-reference limit of §4: it biases that station’s singles by a computable amount, giving an in-situ measurement of the drift susceptibility product $\kappa\lambda$ at the absorbers themselves. The run order is therefore: single-station, fixed phase — calibrate $\kappa\lambda$ from the singles bias; then two-station, phase-locked, shot-randomized global phase — read the correlators. This closes the loophole that would otherwise drain a null result of meaning (“the injection never reached the absorbers”): with $\kappa\lambda$ independently measured, a two-station null bounds the *mechanism*, not the plumbing, and the tone-side prediction becomes falsifiable rather than merely confirmable.

Status and caveats. The tone-side prediction is presented at the same evidentiary level as §8.6(iv)–(v): the injected tone is modeled in the single-quantum sector as a share drift, whereas a real coherent field adds quanta and ultimately requires the multi-quantum machinery of §6.3; the ψ -invariance of δS at the standard angles is a leading-order statement (simulation shows comparable second-order response-curvature corrections at the tested $\lambda = 0.3$, with δS positive throughout); and near-simultaneous commits — within any finite commit duration the foliation cannot order — would escape conditioning at a rate that *grows* with cross-station correlation, a channel we flag as open. The broadband knob carries none of these caveats: the protection theorem and the leak law use only the §7 construction itself, and Knob 1 is proposed as a near-term, systematics-robust measurement of the one parameter (η) on which the entire entangled sector turns.

9. Discussion

9.1 In what sense is this a derivation of the Born rule?

Three answers, in decreasing strength of the question. In the strongest sense — probability extracted from a theory containing no probabilistic ingredient — it is not a derivation, and nothing ever will be: something must break ties, and every program in the literature derives the Born rule *from premises*, the real competition being over how few, how physical, and how independently motivated the premises are. In the sense that

we contend matters, it is one: the Born rule here is a *theorem* from premises that are physical statements about detectors, none of which mentions outcome probabilities. The squaring enters as oscillator energetics (P1); the randomness enters as ordinary noise (P2–P3); their conjunction is forced onto the unique fair point by proofs, not tuning; and the quantity $|A|^2$ is nowhere assigned a probabilistic role by hand. And in one specific respect the mechanism is *stronger* than the axiom it replaces: it specifies the physical conditions under which Born statistics fail (§8). An axiom cannot be wrong in instructive ways; a mechanism can, and twice in this work the mechanism’s own theorems eliminated deviation channels that our first analysis had left open — a pattern we have displayed rather than hidden (Table 1).

9.2 Relation to other programs

Table 2 — Born-rule derivation programs compared.

Program	Selection mechanism	Noise physically identified	Fairness / measure status	Born statistics	Modifies quantum dynamics	Predicts deviations
Gleason (1957)	none (uniqueness theorem)	—	—	unique consistent measure	no	no
Deutsch–Wallace	none (rational credence)	—	—	derived as credence	no	no
Zurek envariance	none (symmetry argument)	—	—	derived from symmetry	no	no
Dürr–Goldstein–Zanghi	outcomes from beables; probabilities from typicality	—	equivariant measure, postulated typical	exact in equilibrium	no (adds beables)	no
Valentini	relaxation of the measure	—	equilibration dynamics	equilibrium value	no (adds beables)	yes (non-equilibrium)
Pearle / GRW / CSL	stochastic reduction	no	postulated (engineered noise)	exact	yes (new constants)	yes (heating, X-ray)
Hughston / ABBH	stochastic reduction	no	postulated (martingale by construction)	exact	yes	yes (weak)
Adler trace dynamics	emergent CSL	partially (substrate fluctuations)	postulated at effective level	exact	fundamental-level replacement	as CSL
Khrennikov PCSFT	threshold detection	yes (classical random field)	—	approximate	replaces QM with field theory	yes
Allahverdyan–Balian–Nieuwenhuizen	apparatus phase transition	yes (apparatus bath)	— (weights from QM)	inherited	no	no
This work	detector energy game	yes (vacuum/environmental noise at the detector)	derived (Theorems 1–5)	exact	no	yes (specified family, §8)

Three neighbors deserve individual comment. **Valentini’s** quantum non-equilibrium is the nearest analogue of our deviation structure: there too the Born rule is an equilibrium condition that suitable physics could violate, with relaxation explaining why violations are unobserved. The difference is locus — relaxation of the configuration-space measure there, fairness of a detector-level game here — and consequently the predicted deviation signatures differ entirely (primordial/exotic matter there; engineered detector asymmetries here). **Khrennikov’s** threshold-detection program is the nearest detector-side analogue: classical random fields plus threshold detectors yielding Born statistics. It obtains Born approximately and calibration-dependently, without the winner-take-all competition; the martingale structure is what buys exactness here. **Allahverdyan, Balian and Nieuwenhuizen’s** Curie–Weiss analysis is the nearest apparatus-side analogue: registration as a metastable apparatus tipping into one basin, treated with full statistical-mechanical care within unitary quantum theory. It takes the outcome weights from standard quantum theory; the present work supplies the selection stochastics beneath such weights. We regard the three as complementary flanks of the same position — that measurement is detector physics — approached from the state, the field, and the apparatus respectively; this paper approaches it from the noise.

9.3 Relation to a broader program

The premises of §2 are not free-floating. They are realized in a specific substrate framework — the Dirac–Kuramoto program — in active development by the present authors, which supplied the wave ontology behind P1, the coupling postulate P2, the locking threshold of P4(i), and the shared registry and foliation of P5–P6, and which motivated this investigation. The framework’s development is publicly documented in a version-controlled repository [Framework repository], whose complete, timestamped revision history constitutes the provenance record of the program, including the evolution of the present mechanism through its intermediate (and twice-retracted) forms. A companion paper develops the measurement-structure consequences — the classical–quantum boundary as the locking threshold quantified in Theorem 2. The present paper is nonetheless deliberately self-contained: its results are conditional on the stated premises and stand or fall with them, not with the framework; any ontology supplying P1–P6 inherits every theorem.

9.4 Open problems

- (i) *The fork of §8.5*: whether a deeper fairness principle closes the mismatched-port channel — the mechanism’s most consequential open question, and experimentally addressable (§8.4).
- (ii) *Deriving $\eta = 1$* : P5’s faithful renormalization is measured to hold at the percent level; the frame-rotation picture of §7.3 suggests why it should be exact, but a dynamical derivation is lacking.
- (iii) *Multipartite extension*: the GHZ configuration, where the two-stage construction must iterate.
- (iv) *Deriving P1 microscopically* within a concrete relativistic oscillator model, replacing generic energetics; and promoting the interference-cross-term origin of P2’s noise scaling (§4) from scaling argument to a controlled open-system calculation — with conservation enforced and the companion terms (heating, additive floor, damping) tracked — which would make the fair noise a *result* of surface wave mechanics rather than a premise about it, with the fairness window $\hbar\omega \gg k_B T$ of §4 expected to emerge as the calculation’s domain of validity.
- (v) *The dressed-mode comparison* for radiative-dominated arrays, required before the §8.3 channel’s magnitude can be trusted.
- (vi) *Formalizations*: general- N induction to continuous-mode Glauber theory; arrays mixing detector classes; second-order corrections (slaved-site correlations; the small equalizing residual in strongly correlated matched ports).
- (vii) *The injection test’s open edges* (§8.7): the multi-quantum treatment of the phase-locked tone; the finite-commit-duration ordering window, whose escape rate grows with cross-station correlation; and the small- λ regime where the leading-order tone law becomes quantitatively clean.
- (viii) *The return-rate ansatz* (escalated from the companion measurement-structure paper’s review panel, finding N2): derive $\kappa_{\text{ret}} = \Delta E/\hbar$ — and the legitimacy of modeling reversible give-back as a contraction eigenvalue in the site’s reduced description — from an explicit $H = H_{\text{field}} + H_{\text{absorber}} + H_{\text{bath}} + H_{\text{int}}$, identifying the actual reduced-dynamics poles; and settle, in the same calculation, whether vacuum coupling restores an effective free phase below the classical threshold.

The one-sentence summary of the paper is this: with the noise physically identified and the fairness proved rather than postulated, the Born rule is what remains of the measurement postulate — and what remains is

bookkeeping: a fair fight over deposited energy, in one world, with the odds set by mechanics and the fights we could rig identified in advance.

Appendix A — Share drift under general noise: proofs

A.1 General noise law, independent sites. Let $de_i = \sigma a(e_i) dW_i$ with independent Wiener increments. Itô's lemma on $s_i = e_i/E$, using $\partial s_i / \partial e_j = \delta_{ij}/E - e_i/E^2$ and $\partial^2 s_i / \partial e_j^2 = -2\delta_{ij}/E^2 + 2e_i/E^3$, gives

$$\mathbb{E}[ds_i] = \frac{\sigma^2}{2} \sum_j a(e_j)^2 \frac{\partial^2 s_i}{\partial e_j^2} dt = \frac{\sigma^2}{E^2} \left[s_i \sum_j a(e_j)^2 - a(e_i)^2 \right] dt.$$

This vanishes for *all* configurations iff $a(e)^2 \propto e$ — the amplitude-linear law — proving Theorem 1 and its uniqueness. The two failure laws quoted in §5.1 are immediate: $a = a_0$ (additive) gives drift $(\sigma^2 a_0^2 / E^2)(Ns_i - 1)$, positive for above-average sites; $a = e$ (multiplicative) gives $(\sigma^2 e_i / E^2)(\frac{1}{E} \sum_j e_j^2 - e_i)$, negative for the leader.

A.2 Correlated noise. With $a(e) = \sqrt{e}$ and $\langle dW_i dW_j \rangle = C_{ij} dt$, the mixed second-derivative terms $\partial^2 s_i / \partial e_j \partial e_k = -(\delta_{ij} + \delta_{ik})/E^2 + 2e_i/E^3$ contribute and the same computation yields the formula of Theorem 3:

$$\mathbb{E}[\dot{s}_i] = \frac{\sigma^2}{E^2} \left[s_i (\sqrt{e} \cdot C \cdot \sqrt{e}) - \sqrt{e_i} (C \sqrt{e})_i \right].$$

Checks: $C = \mathbb{1}$ recovers A.1's zero; $C_{jk} \equiv 1$ (common noise) gives, for $N = 2$, $\mathbb{E}[\dot{s}_1] = (\sigma^2/E^3) \sqrt{e_1 e_2} (e_1 - e_2)$ — the rich-get-richer drift quoted in §5.3; linearity in C underlies the mixed-kernel scaling (deviation $\propto f$) exploited in §5.3 and §8.

Appendix B — The slaved phase: linear analysis and layer width

B.1 The sub-threshold mode. In a frame rotating at the drive frequency, with the detuning convention $\dot{a} = -(\kappa_{\text{ret}} + i\Delta)a + f e^{i\varphi_d}$ and $\kappa_{\text{ret}} = \Delta E / \hbar$ the off-shell return rate (§5.2 dictionary) (P1, P4(i)), the system is linear and inhomogeneous. Its unique fixed point is $a^* = f e^{i\varphi_d} / (\kappa_{\text{ret}} + i\Delta)$, i.e. $r^* = f / \sqrt{\kappa_{\text{ret}}^2 + \Delta^2}$, $\varphi^* = \varphi_d - \arctan(\Delta / \kappa_{\text{ret}})$; the homogeneous eigenvalues are $\lambda = -\kappa_{\text{ret}} \pm i\Delta$, so the flow is a global contraction. No trajectory winds: phase deviations decay at rate κ_{ret} , and the stationary phase offset is a function of $(\kappa_{\text{ret}}, \Delta)$ *only* — identical for identical sites, independent of stored energy. By contrast, the Adler equation $\dot{\varphi} = \Delta - K \sin \varphi$ presupposes an *autonomous* oscillator: amplitude frozen at a limit-cycle radius maintained by the oscillator's own gain, phase a neutrally stable variable free to run. Its running solutions (for $|\Delta| > K$) carry the nonzero average $\langle \sin \varphi \rangle = [\Delta - \text{sgn}(\Delta) \sqrt{\Delta^2 - K^2}] / K$ — the entrainment pull. Below threshold there is no gain, no limit cycle, no running solutions, and hence no pull: the rich-get-richer channel is structurally absent, which is Theorem 2.

B.2 Layer width, two ways. (i) *Marginality*: the phase becomes exploitable when its restoring rate falls below the coupling, $\kappa_{\text{ret}} \lesssim K$, i.e. $\Delta E \lesssim \hbar K$, giving fractional width $w = K/\omega$. (ii) *Width-commitment identity*: the same Γ that broadens the final state is, by fluctuation-dissipation, the rate of irreversible decay into registered products; with $K \sim \Gamma$ this reproduces $w = \Gamma/\omega$. The two estimates agree because they are the same physics seen from the dynamical and the dissipative side.

B.3 Feedback hierarchy. From $\dot{e} = 2r\dot{r} = -2\kappa_{\text{ret}} e + 2f\sqrt{e} \cos(\varphi_d - \varphi)$: uptake scales as $e^{1/2}$, decay as e^1 . For two sites exchanging via any channel with uptake $u(e) = c e^p$, the share drift obeys $\dot{s}_1 \propto u(e_1)e_2 - u(e_2)e_1 = e_1 e_2 [u(e_1)/e_1 - u(e_2)/e_2]$, whose sign for $e_1 > e_2$ is that of $p - 1$: concave uptake ($p < 1$) equalizes, linear ($p = 1$) is share-neutral, superlinear ($p > 1$) amplifies. The slaved sector contains only $p = \frac{1}{2}$ (equalizing, toward the drive pattern — which by P1 is the Born pattern) and $p = 1$ (neutral); the amplifying sector requires autonomy and is confined to the layer.

Appendix C — Kinematic linearity of registration (Theorem 5)

Registration is a transition from the delocalized n -quantum excitation to a final state containing a registered excitation of energy $\hbar \omega$ at site i (plus $n - k$ quanta remaining). At leading order in the registration

coupling, the T-matrix element consists of k absorption vertices at the site; each vertex is amplitude-linear (P2), so $M_i \propto \psi^{(k)}(i, \dots, i)$ — one factor of the k -quantum joint amplitude — and the golden rule gives $\text{rate}_i \propto |\psi^{(k)}(i, \dots, i)|^2 = G^{(k)}(i)$: linear in the coincidence weight, never a higher power. The alternatives are eliminated as follows. *Fewer than k vertices*: the energy deficit has no compensating final state (P4(i): no sub-gap levels), and thermal compensation is Boltzmann-suppressed ($e^{-\Delta E/k_B T} \sim e^{-10}$ already at $\Delta E = 0.25$ eV, room temperature). *More absorption vertices*: $(k+1)\hbar\omega$ final states are absent in-sector, or constitute a different channel, itself linear in its own $G^{(k+1)}$. *Absorb-emit insertions*: return quanta to the pool — they are the exchange dynamics of §4, not registration; their leading contribution to the commit channel is a three-vertex path whose amplitude is suppressed by two powers of (coupling/energy denominator), a parametric estimate $(\Gamma/\omega)^2 = w^2 \lesssim 10^{-4}$ for broadband silicon and $\lesssim 10^{-14}$ for atomic lines. *Saturation*: occupation factors $(1 - n_f)$ equal unity for a ground-state detector. Combining with the affine ensemble structure (§6.3): every commit channel is linear in a $G^{(k)}$, every $G^{(k)}$ is linear in ρ , per-configuration statistics are Born (§5), and mixtures average affinely — all outcome statistics are affine in ρ , in every sector.

Appendix D — Simulation methods and code availability

All simulations use a common stochastic engine: Euler–Maruyama integration of $de_i = \sigma a(e_i) \Delta W_i$ with $\Delta t = 0.02$, $\sigma = 0.3$ (units $E_0 = 1$), and clipping at zero (for the $\sqrt{e}$ law, zero is naturally sticky — a CIR-type boundary — implementing drop-out); the discrete gambler variants use stakes-scaled exchange $\delta = \pm \text{step} \cdot \min(e_i, e_j)$ with fair sign. Registration is implemented either as a share threshold (absorption at $s \geq 0.7$ – 0.995 as noted per experiment) or as a rate process (geometrically distributed commit times; conditional pick $\propto s^\alpha$). Correlated noise uses covariance factorization by eigendecomposition with eigenvalue clipping. Ensemble sizes are 1.5×10^3 – 6×10^4 trials per data point, with Monte-Carlo floors quoted alongside results in the text. Deliberately *failing* configurations (clipped fixed-step exchange; additive and multiplicative noise laws; extreme-value selection; biased-layer profiles; common-mode noise; nonlinear commit rates) are retained in the code base as diagnostics: each failure realizes a specific theorem’s converse, and their outputs are reported in §§4–8 with the same prominence as the passing runs. The complete simulation suite (thirteen scripts, all random seeds pinned, outputs matching every number quoted in this paper) is archived with an index mapping each script to the results and figures it supports [repository ref]. The two-station injection-test simulation of §8.7 (correlated Wright–Fisher joint games with the §7.1 conditional reset; same-step double commits ordered explicitly, without which correlated paths silently skip conditioning and mimic η -degradation) is archived separately as `two_station_sims/` with its full-statistics output.

Appendix E — Closed forms for hypothetical deviations

E.1 Signaling amplitude of a commit nonlinearity (§7.4). For fast commitment with pick probability $\propto s^{1+\varepsilon}$ at wing A , state $\sqrt{p}|00\rangle + \sqrt{1-p}|11\rangle$, Alice at angle α : her local shares are $s_+ = p \cos^2 \alpha + (1-p) \sin^2 \alpha$, $s_- = 1 - s_+$, and Bob’s conditionals are $P(B=0 | A=\pm) = p \cos^2 \alpha / s_+$, $p \sin^2 \alpha / s_-$. Composing,

$$M(B=0; \alpha) = p \frac{\cos^2 \alpha \frac{s_+^\varepsilon}{s_+^{1+\varepsilon}} + \sin^2 \alpha \frac{s_-^\varepsilon}{s_-^{1+\varepsilon}}}{\frac{s_+^\varepsilon}{s_+^{1+\varepsilon}} + \frac{s_-^\varepsilon}{s_-^{1+\varepsilon}}},$$

which is α -independent ($= p$) iff $\varepsilon = 0$. Expanding at $\alpha = 0$ against the fixed point $M(45^\circ) = p$:

$$\Delta M = \varepsilon p(1-p) \ln \frac{p}{1-p} + O(\varepsilon^2), \quad K(p) \equiv p(1-p) \ln \frac{p}{1-p},$$

with $K(0.8) = 0.222$ at leading order (effective $K \approx 0.18$ – 0.21 over the simulated ε range), halved in the full process by $P_{\text{first}} = \frac{1}{2}$. Both exact zeros — $\varepsilon = 0$ and $p = \frac{1}{2}$ — are manifest in the formula.

E.2 Port-level drift and the κ model (§8.2–8.3). Summing A.2 over the sites of port A (block-structured C ; cross-port correlations zero) gives

$$\dot{S}_A = \frac{\sigma^2}{E^2} (q_B S_A - q_A S_B), \quad q_P = \sum_{i,j \in P} C_{ij} \sqrt{e_i e_j},$$

vanishing iff $q_A/S_A = q_B/S_B$ (matched collective structure). For the uniform-port model — m equal-energy sites per port, internal correlation c_P — one finds $q_P = S_P E [1 + (m-1)c_P]$ and hence

$$\dot{S}_A = \frac{\sigma^2}{E} S_A S_B (m-1) (c_B - c_A) :$$

drift proportional to the collective-structure mismatch, suppressing the internally-correlated port, and vanishing for matched ports at any c . A constant drift-to-diffusion ratio integrates to the biased-ruin form $P'(S) \approx S + \kappa S(1-S)$ quoted in §8.3, with $\kappa \propto (m-1)(c_B - c_A)$; the simulations give effective $\kappa \approx -\Delta c_{\text{eff}}$ at $m = 4$ over the tested range.

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